

HKS ATTENDANCE
(HYBRID KNOWLEDGE SYSTEM
ATTENDANCE)
A SECURE MULTI-EVIDENCE ATTENDANCE
VERIFICATION PLATFORM

*Built on Bluetooth Low Energy Proximity Sensing, GPS Geofencing, Device Binding,
Faculty Governance, and Claims-Based Dispute Resolution*

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Submitted by

Penumarthi Navya Sree Ram Kumar Chowdary

Roll No.: CS21B1039

(Batch 2022 – 2026)

Under the Guidance of

Dr. Harish Kumar Sardana

Professor, Computer Science & Engineering Department, IIIT Raichur

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY
RAICHUR-584135, INDIA**

Academic Year 2025 – 2026

DECLARATION

I hereby declare that the project entitled “*HKS ATTENDANCE – Hybrid Knowledge System Attendance*“, submitted by me, is an original piece of work carried out under the supervision of Dr. Harish Kumar Sardana, and that it has not been submitted, in whole or in part, to any other institution for the award of any degree or diploma.

Raichur – 584135
Academic Year 2025 – 2026

Penumarthi Navya Sree Ram Kumar
Chowdary

CERTIFICATE

This is to certify that the work contained in this project report entitled “*HKS ATTENDANCE – Hybrid Knowledge System Attendance*”, submitted by **Penumarthi Navya Sree Ram Kumar Chowdary** in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is a bona fide record of work carried out under my supervision and that it has not been submitted elsewhere for the award of any degree.

Dr. Harish Kumar Sardana
Project Supervisor

Head of Department
Department of Computer Science and
Engineering

Department Seal

Raichur – 584135
Academic Year 2025 – 2026

(Dr. Harish Kumar Sardana)
Project Supervisor

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ABSTRACT

Conventional attendance-recording methods, including manual registers and single-channel digital systems such as RFID cards, scannable codes, or GPS-only checks, share a common weakness: each relies on one piece of evidence that can be transferred, forged, or spoofed. This project presents HKS ATTENDANCE (Hybrid Knowledge System Attendance), an attendance-verification platform designed on the principle that no single evidence source should determine an attendance outcome on its own.

The system requires a student’s device to corroborate presence through two independent channels: a Bluetooth Low Energy (BLE) handshake with a beacon installed in the classroom, and a GPS reading validated against that classroom’s geofence. Each student account is bound to a single registered device, preventing credential sharing without administrative intervention. Submissions for which the two evidence channels disagree are flagged for review by the course faculty member rather than being resolved automatically. The faculty member can inspect the underlying evidence – signal strength, beacon identity, coordinates, and distance from the room – before confirming or rejecting the record. Students may contest a rejected outcome through a structured claims workflow, with the owning faculty member adjudicating the dispute and the resolution permanently logged.

The platform was implemented as a complete working system: a Flutter application for students and staff, a FastAPI backend exposing a versioned REST interface, a PostgreSQL database enforcing referential integrity across the academic and security entities, and ESP32 microcontrollers reprogrammed as classroom beacons. Each beacon payload is protected with a single-use nonce and an HMAC signature to resist replay attacks, and every governance action taken by a faculty member or administrator is written to an immutable audit log.

The completed system was evaluated end to end across authentication, device registration, beacon discovery, location validation, attendance decisioning, faculty review, administrative governance, and claim resolution. The results indicate that combining two independently weak, independently spoofable signals into a single decision produces an attendance system that is more resistant to manipulation than either signal used alone, while keeping the everyday experience for students and staff close to existing digital attendance workflows.

In summary, the implemented system successfully integrates BLE proximity verification, GPS geofence validation, device binding, faculty governance, claims-based dispute resolution, audit logging, and cloud deployment into a single, deployable attendance plat-

form. Validation across authentication, attendance submission, governance workflows, and a deliberate set of security misuse scenarios demonstrates the practical feasibility of the proposed multi-evidence architecture, and indicates that it is meaningfully more resistant to the impersonation and single-channel weaknesses identified in the existing literature than the manual, card-based, and single-signal systems still in widespread use. The architecture has also been designed with extensibility in mind, allowing future integration of indoor localisation algorithms and computer-vision-based classroom validation.

Keywords: Attendance Management, Bluetooth Low Energy, BLE Beacon, GPS Geofencing, Device Binding, FastAPI, Flutter, PostgreSQL, Replay Protection, Faculty Governance, Claims Resolution, Multi-Evidence Verification.

LIST OF ABBREVIATIONS

API	Application Programming Interface
BLE	Bluetooth Low Energy
CSV	Comma Separated Values
GPS	Global Positioning System
HKS	Hybrid Knowledge System
HMAC	Hash-Based Message Authentication Code
HTTP	HyperText Transfer Protocol
JWT	JSON Web Token
ORM	Object Relational Mapper
REST	Representational State Transfer
RSSI	Received Signal Strength Indicator
SQL	Structured Query Language
UI	User Interface
UUID	Universally Unique Identifier

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Chapter 1

Introduction

1.1 Background

Attendance recording carries more administrative weight than its routine appearance suggests. It feeds into how regularly a student is judged to have engaged with a course, into the data an institution reports to accrediting and regulatory bodies, and into the everyday decisions faculty make about pacing and academic support. For most of the history of formal education, this responsibility was carried by a paper register and a roll call – a method that requires little of either students or staff but offers almost no protection against misuse: a classmate can answer for someone who never arrived, a page can be misplaced, and there is no record after the fact of why a particular mark was made.

As institutions digitised their administrative processes, attendance was an early target. Card-based readers, biometric scanners, and location-aware mobile applications have each been deployed at different points, and each has reduced some of the friction associated with manual recording. However, most of these systems share a reliance on a single channel of evidence to decide whether a student was present. A card can be lent to another student. A face can occasionally be substituted by a photograph under poor lighting. A phone’s reported location can be inaccurate indoors or deliberately falsified using a mock-location tool. None of these weaknesses is fatal on its own, but each is significant enough that an attendance system built around a single channel inherits that channel’s weakness in full.

This thesis presents the design, implementation, and evaluation of HKS ATTENDANCE, an attendance platform built on the principle that attendance decisions should be derived from corroborated, multi-channel evidence rather than a single indicator. The system combines short-range Bluetooth proximity sensing with device GPS, together with a binding between a student’s account and a single physical device, with the goal of producing attendance decisions that are harder to manipulate than those produced by any single signal, without materially increasing the effort required from students or staff.

1.2 Problem Statement

Attendance workflows in widespread use today exhibit a recurring set of weaknesses:

- Reliance on a single verification channel, so that defeating one mechanism is sufficient to defeat the entire system.
- Limited or no mechanism for a student to contest an attendance outcome believed to be incorrect.
- Sparse audit trails, so institutions cannot later reconstruct why a particular record was accepted or rejected.
- Little resistance to one person submitting attendance on behalf of another.
- Weak protection against the replay of previously captured verification data.
- Administrative tooling that is added after the fact rather than designed alongside the attendance logic itself.

This project addresses these weaknesses by designing and implementing a platform in which attendance decisions are derived from corroborated, independently verifiable evidence, with faculty oversight, administrative governance, and a structured appeals process designed in from the outset.

1.3 Motivation

No single sensing technology available on an ordinary smartphone can, by itself, establish that its owner is physically present in a particular classroom. Satellite positioning degrades indoors and can be manipulated through a mock-location utility. Short-range radio proximity can confirm that a phone is near a fixed point but says nothing about which account is operating that phone, or whether the same credentials are simultaneously in use on a second device elsewhere. Manual oversight by an instructor does not scale to large sections and is itself susceptible to error.

Early planning for this project established that no individual technology would be sufficient on its own, and that the more promising direction was to make multiple imperfect signals corroborate one another. A phone that demonstrates both BLE proximity to a classroom beacon and GPS coordinates within that classroom's geofence provides

considerably stronger evidence than a phone that satisfies only one of these conditions. A one-student-one-device policy closes off the simplest remaining avenue for impersonation: sharing login credentials.

1.4 Objectives

The project set out to achieve the following objectives:

1. Design an attendance architecture that bases its decisions on more than one independent evidence source.
2. Implement Bluetooth Low Energy proximity verification using purpose-built classroom beacons.
3. Implement GPS-based geofence validation as a secondary, corroborating evidence source.
4. Bind each student account to a single authorised device to prevent credential sharing.
5. Build a faculty-facing review workflow for attendance records flagged as inconclusive by the decision engine.
6. Build an administrative layer for managing the academic structures – users, courses, enrolments, classrooms, and beacons.
7. Harden the evidence-collection pipeline against replay and forgery through nonce validation and cryptographic signatures.
8. Provide students with a structured mechanism to contest attendance outcomes considered incorrect.
9. Produce an architecture that is suitable for pilot deployment in a real academic environment and extensible toward institution-scale adoption.

1.5 Scope of the Project

The implemented system covers:

- Account authentication and attendance submission for students.

- Collection of BLE proximity evidence from beacons registered as trusted by the institution.
- Collection and validation of GPS evidence against classroom geofences.
- Registration and enforcement of device bindings.
- Discovery of active attendance sessions scoped to a student’s enrolments.
- Faculty-side review and resolution of attendance records requiring manual attention.
- Administrative management of the academic hierarchy and BLE infrastructure.
- Submission and resolution of attendance claims.
- Security validation, including replay protection, and audit logging of governance actions.

The following were deliberately treated as out of scope for the current iteration of the project:

- Facial recognition or other camera-based biometric verification.
- Offline attendance capture with deferred synchronisation.
- Predictive analytics or machine-learning-driven risk scoring.
- Deployment across multiple, independently administered institutions.
- Federation of the platform across multiple campuses of the same institution.

For ease of reference, Table 1.1 summarises the complete technology stack used to build HKS ATTENDANCE; each choice is revisited and justified in detail in Chapters 3 and 4.

1.6 Contributions

This work makes the following contributions:

- A working hybrid attendance-verification architecture combining BLE and GPS evidence.

Table 1.1: Technology Snapshot

Layer	Technology
Mobile Application	Flutter 3.x with Riverpod
Backend Framework	FastAPI (Python) with Pydantic
Database	PostgreSQL
ORM / Migrations	SQLAlchemy / Alembic
Authentication	JSON Web Tokens (JWT)
BLE Hardware	ESP32 microcontrollers
Beacon Payload Security	HMAC-SHA256

- A device-binding scheme that ties a student’s attendance privileges to a single registered device.
- A faculty governance workflow built around evidence inspection rather than fully automated decisioning.
- An administrative layer covering the full academic hierarchy and the BLE beacon estate.
- A claims and dispute-resolution workflow with a permanent audit trail.
- A nonce-based replay-protection scheme for beacon broadcasts.
- A comprehensive audit-logging design covering every governance decision in the system.
- An end-to-end implementation spanning a mobile client, backend services, and embedded beacon firmware.

1.7 Research Methodology

The project followed a sequential research methodology, illustrated in Figure 1.2, moving from problem identification through to evaluation rather than treating design and implementation as a single undifferentiated activity.

The work began with problem identification: an examination of how attendance is recorded in practice and why existing digital systems continue to be defeated by simple impersonation. This was followed by a literature review covering manual, card-based, code-based, satellite-positioning, biometric, BLE-based, and camera-based approaches,

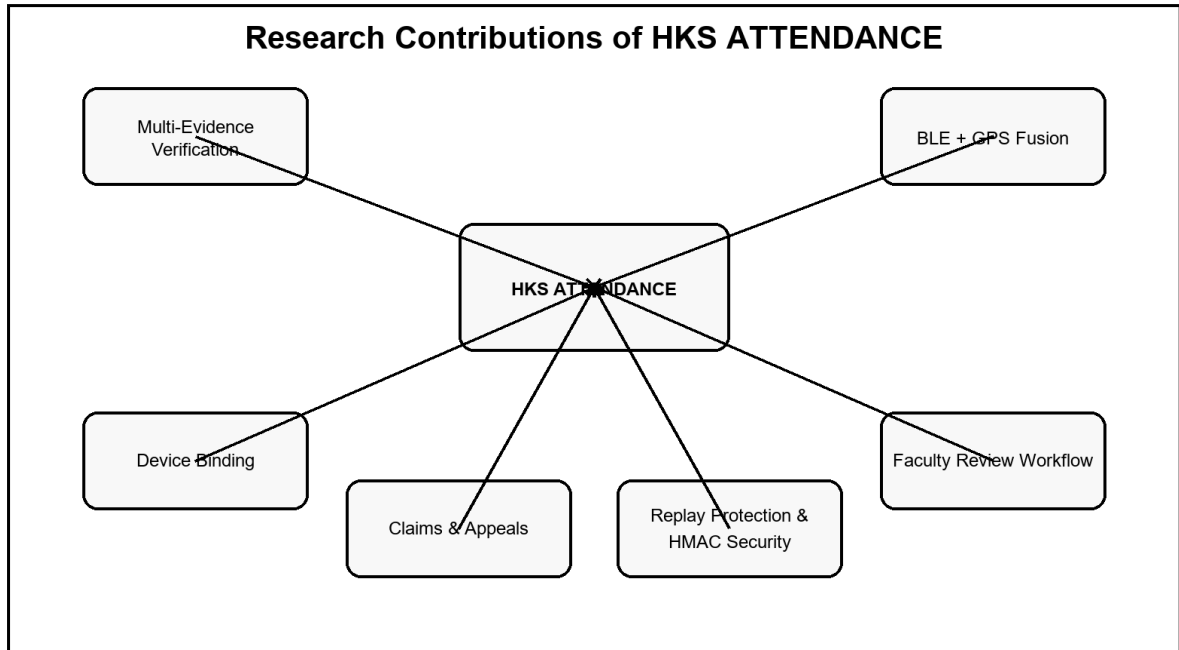


Figure 1.1: Research Contributions of HKS ATTENDANCE

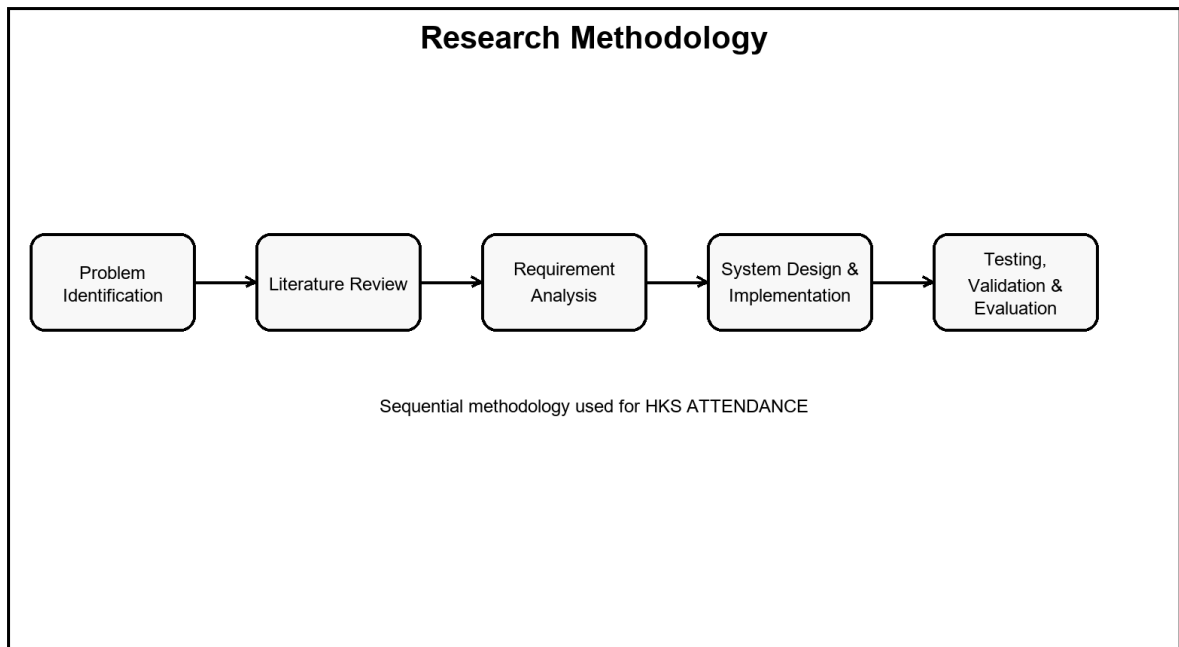


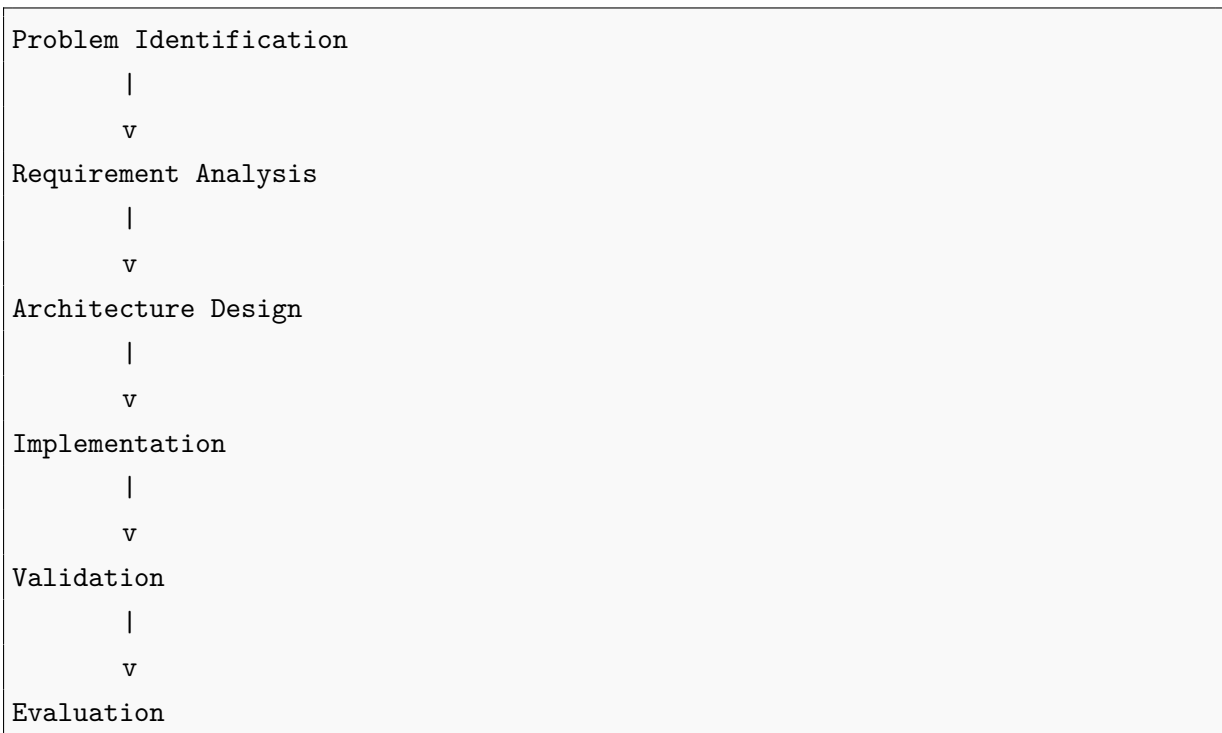
Figure 1.2: Research Methodology

summarised in Chapter 2. The literature review fed directly into a research gap analysis, which identified that very few existing systems combine more than one verification channel, almost none bind a student account to a specific device, and fewer still provide a structured path for faculty review or student appeal.

These gaps shaped the requirement analysis presented in Chapter 3, which translated the identified weaknesses into functional and non-functional requirements. BLE and GPS were selected as the two corroborating evidence channels because they are independently available on commodity smartphones, fail in different and largely unrelated ways, and can be combined without specialised hardware beyond a low-cost beacon. System design (Chapter 4) and database design (Chapter 5) followed, translating these requirements into a layered architecture and a normalised relational schema. Implementation (Chapter 6) produced a working Flutter, FastAPI, and PostgreSQL system together with ESP32 beacon firmware. Testing and validation (Chapter 7) then exercised the system across functional, security, and performance dimensions, and the results were evaluated against the original objectives in Chapter 8 before concluding in Chapter 9.

1.7.1 Research Design

Figure 1.3 restates the methodology of Section 1.7 as an explicit research-design pipeline, the form in which this kind of sequential, requirements-driven engineering research is typically presented.



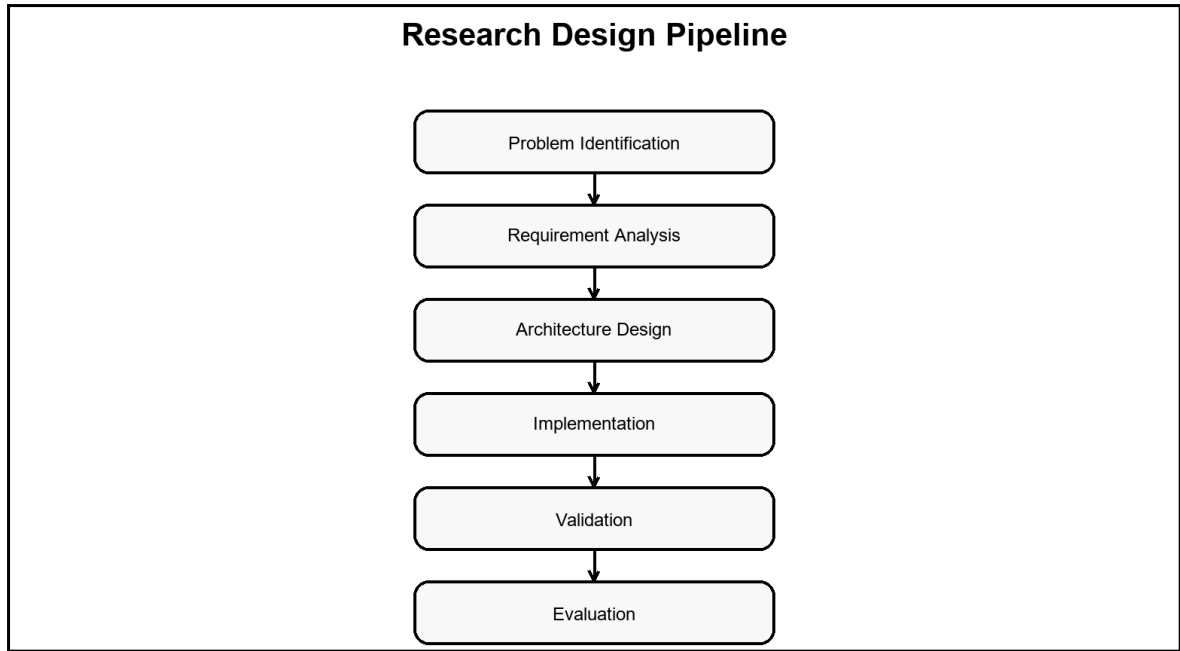


Figure 1.3: Research Design Pipeline

Each stage in this pipeline corresponds directly to a chapter of this thesis: problem identification and requirement analysis are addressed in Chapters 2 and 3, architecture design in Chapters 4 and 5, implementation in Chapter 6, validation in Chapter 7, and evaluation in Chapter 8.

1.8 Organisation of the Thesis

The remainder of this thesis is structured as follows. Chapter 2 surveys existing approaches to attendance automation and positions this project relative to them. Chapter 3 presents the system analysis, including requirements and feasibility. Chapter 4 describes the system architecture and module-level design. Chapter 5 covers the database design. Chapter 6 details the implementation of each subsystem, including deployment architecture and threat modelling. Chapter 7 presents the testing and validation methodology and results. Chapter 8 discusses outcomes, comparative positioning, implementation footprint, and limitations. Chapter 9 concludes the thesis and outlines future work.

1.9 Chapter Summary

This chapter introduced the problem of single-channel attendance verification, the motivation for combining BLE and GPS evidence, the project’s objectives and scope, its principal contributions, the research methodology followed, and the organisation of the remaining chapters. The next chapter presents the literature review that informed the design of HKS ATTENDANCE.

Chapter 2

Literature Review

2.1 Introduction

Before settling on a hybrid BLE-and-GPS architecture, it is necessary to examine the broader landscape of attendance-automation technologies that have been studied and deployed over the past two decades, including those this project deliberately did not build on. This chapter reviews manual registers, card-based systems, code-scanning systems, satellite-positioning systems, biometric systems, short-range radio systems, and camera-based recognition systems; it then reviews a representative set of research papers addressing BLE- and GPS-based attendance verification, before positioning HKS ATTENDANCE relative to the gaps identified. Sections 2.3 through 2.5 and 2.8 describe technologies that do not form part of the system implemented in this project; they are reviewed solely as prior art.

2.2 Manual Attendance Registers

The oldest and still most common approach is for an instructor to call a roll and mark a paper or spreadsheet register by hand. This method requires no infrastructure and no training, but it scales poorly with class size, consumes teaching time, and offers no resistance to a student answering on a classmate's behalf. Manual registers are also difficult to audit: once a mark is made, there is rarely a record of the basis on which it was made, and disputed entries are difficult to revisit months later.

2.3 Card-Based (RFID) Systems

Radio-Frequency Identification readers were among the first automated alternatives to the paper register. Each student is issued a card carrying a unique identifier, and a reader installed at the classroom entrance logs a tap as an attendance event. This reduces manual effort and can process a queue of students quickly, but the underlying weakness is

structural: a card is a bearer credential, and nothing prevents one student from carrying a friend’s card through the reader. Equipping every classroom with reader hardware also adds a recurring infrastructure and maintenance cost that scales with institution size.

2.4 Code-Scanning Systems

A lighter-weight alternative that gained popularity with the spread of smartphones is the display of a time-limited scannable code at the start of a session, which students photograph to register presence. This requires no dedicated hardware beyond a projector and a phone camera. The same lightness, however, undermines it as a verification mechanism: a code displayed on a screen can be photographed and forwarded to anyone, and scanning it does not establish physical presence in the room. Static or long-lived codes are particularly susceptible to this kind of sharing, and even short-lived codes do little to confirm that the scanning device is physically present rather than relying on a forwarded screenshot.

2.5 Satellite-Positioning (GPS-Only) Systems

Systems that rely solely on a device’s reported GPS coordinates check those coordinates against a geographic boundary drawn around the institution or a particular building. This requires no additional hardware, since location services are already present on virtually every smartphone, but satellite positioning was not designed for indoor, room-level precision. Signal degradation inside buildings, multipath reflection in dense surroundings, and the relative ease of installing a mock-location utility on a rooted device combine to make GPS, used alone, a weak basis for a presence claim. It can usually confirm that a phone is somewhere on or near a campus; it struggles to confirm which classroom that phone is in, or whether the reported coordinates are genuine.

2.6 Biometric Systems

Fingerprint, iris, and facial-recognition systems verify identity through a physiological trait rather than a possession (a card) or a secret (a password), making casual impersonation considerably harder. The trade-offs are correspondingly larger: dedicated sensors are expensive to deploy at institutional scale, the collection and storage of biometric templates raises privacy and consent questions, and recognition accuracy is sensitive to

lighting, sensor quality, and changes in appearance. These costs have kept biometric attendance from becoming the default choice outside higher-stakes or smaller-scale deployments.

2.7 Bluetooth Low Energy Proximity Systems

Bluetooth Low Energy beacons broadcast a short, periodic radio signal that nearby devices can detect and use to estimate proximity. Because BLE is short-range, it performs better than satellite positioning at distinguishing one room from an adjacent one, while drawing very little power, allowing a beacon to run for months on simple hardware. Used alone, however, BLE has its own gaps: received signal strength is noisy and fluctuates with body position and obstruction, an unauthorised device could imitate a legitimate beacon’s identifier, and a captured broadcast could be recorded and replayed later unless the protocol explicitly guards against it. These observations directly motivated the security hardening – nonce validation, nonce caching, and signature verification – built into the BLE evidence pipeline described in Chapters 4 and 6.

2.8 Camera-Based Facial Recognition Systems

Advances in computer vision have made passive, camera-driven facial recognition a candidate for attendance automation: a camera mounted in a classroom can, in principle, log presence without requiring any action from the student. The cost of that convenience is a substantial compute and camera-infrastructure requirement, sensitivity to lighting and occlusion, and a more pronounced version of the privacy concerns already associated with biometric systems generally, since continuous video capture of a room is a larger intrusion than a single fingerprint scan at the door.

2.9 Hybrid and Multi-Evidence Approaches

A recurring theme in recent research is that no single sensing modality is individually sufficient, and that pairing two complementary, independently weak signals tends to outperform either signal alone. Reported combinations include radio proximity paired with satellite positioning, and biometric verification paired with location checks. The reported benefits are consistent: an attacker must defeat two independent mechanisms rather than one, and the resulting evidence trail is richer, which makes after-the-fact

review and dispute resolution more tractable. The corresponding costs are increased architectural complexity, the need to reconcile evidence arriving through different channels, and a greater governance burden, since a hybrid system inevitably produces ambiguous cases that a single-signal system would simply accept or reject outright. HKS ATTENDANCE adopts this hybrid approach directly, pairing BLE proximity evidence with GPS geofence evidence and routing ambiguous cases to a human reviewer rather than resolving them automatically.

2.10 Comparison with Commercial Attendance Systems

In addition to the academic literature reviewed in Sections 2.3 through 2.9, it is useful to position HKS ATTENDANCE against the commercial and off-the-shelf attendance products that institutions encounter in practice. Table 2.1 summarises five commonly deployed commercial product categories – RFID-based attendance terminals, QR/scannable-code attendance apps, consumer GPS-geofencing attendance apps, commercial face-recognition attendance kiosks, and standalone BLE beacon attendance apps – against the same capability set used in the research gap analysis of Section 2.12.

Commercial RFID terminals and face-recognition kiosks both achieve good room-level precision, but at a hardware cost and (for face recognition) a privacy cost that is difficult to justify for general classroom use. Commercial QR and GPS apps minimise hardware cost but reintroduce the sharing and spoofing weaknesses discussed in Sections 2.4 and 2.6. None of the commercial categories surveyed combine two independent evidence channels, bind a student to a specific device, or expose a faculty-facing evidence-review and appeals workflow – the same gap identified against the academic literature in Section 2.12, reinforcing that this is a gap in current practice generally and not only in the research literature.

2.11 Summary of Literature Findings

Before turning to a formal gap analysis, it is useful to state plainly what the literature reviewed in Sections 2.2 through 2.10 collectively shows. Five findings recur across the sources surveyed:

1. **Finding 1** – Most deployed attendance systems rely on a single verification method,

whether a card, a code, a location reading, or a biometric trait, and inherit that method’s specific weakness wholesale.

2. **Finding 2** – BLE provides meaningfully better indoor proximity resolution than GPS, but is itself vulnerable to signal noise, beacon spoofing, and replay unless explicitly hardened.
3. **Finding 3** – GPS alone cannot reliably distinguish one classroom from an adjacent one, and is comparatively easy to falsify on a rooted or jailbroken device.
4. **Finding 4** – Faculty governance – a human reviewer inspecting raw evidence before a decision is finalised – is rarely discussed or implemented in the systems reviewed, academic or commercial.
5. **Finding 5** – Device binding, in the sense of tying a student’s attendance privileges to one specific registered device, is almost entirely absent from the literature, despite credential sharing being a frequently cited weakness.

These five findings, taken together, motivate the research gap analysis that follows.

2.12 Evolution of Attendance Systems

Figure 2.1 places the technologies reviewed in this chapter on a rough timeline, showing the progression from manual registers through card- and code-based systems to location- and proximity-aware approaches, culminating in the multi-evidence design adopted by HKS ATTENDANCE.

2.13 Research Gap Analysis

Table 2.2 summarises the principal weakness associated with each approach reviewed in Sections 2.2 through 2.8.

Table 2.2 compares technologies by their single principal weakness. Table 2.3 extends this into a broader, multi-feature comparison across the eleven capabilities most relevant to this thesis, making explicit just how many of these capabilities are absent from each existing approach.

The pattern in Table 2.3 is consistent with Table 2.2: existing single-channel technologies each cover at most two or three of the eleven capabilities considered, while HKS ATTENDANCE was deliberately designed to cover all eleven simultaneously.

Table 2.1: Comparison with Commercial Attendance Systems

Capability	RFID	QR	GPS	Face Rec.	BLE
Hardware cost	High (reader per room)	Low	None	Very high (camera + compute)	Low (beacon per room)
Resistant to credential sharing	No	No	No	Partially	No
Resistant to replay	N/A	No	No	N/A	Rarely
Indoor room-level precision	Yes	N/A	No	Yes	Yes
Privacy footprint	Low	Low	Moderate	High	Low
Faculty review of evidence	Rarely	Rarely	Rarely	Rarely	Rarely
Dispute / appeal workflow	No	No	No	No	No

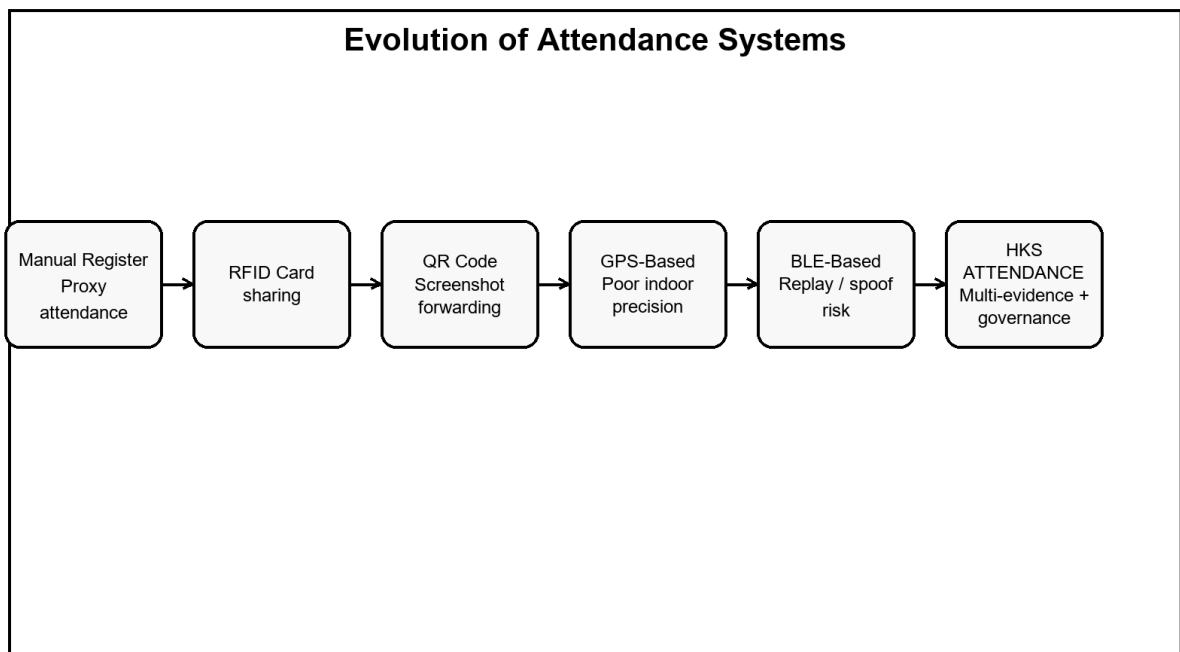


Figure 2.1: Evolution of Attendance Systems

Table 2.2: Principal Limitation by Attendance Technology

Technology	Principal Limitation
Manual Register	No resistance to one student answering for another
RFID Card	Cards are freely transferable between students
Scannable Code	Codes can be photographed and forwarded remotely
GPS Only	Poor indoor precision; coordinates can be falsified
Biometric	High deployment cost; privacy and consent concerns
BLE Only	Signal noise; vulnerable to beacon spoofing and replay
Facial Recognition	Camera infrastructure cost; privacy intrusion

Table 2.3: Comparison Between Existing Systems and HKS ATTENDANCE

Feature	RFID	QR	GPS	BLE	HKS
Device Binding	No	No	No	No	Yes
Faculty Review	No	No	No	No	Yes
Claims / Appeals	No	No	No	No	Yes
Audit Logging	No	No	No	No	Yes
Replay Protection	N/A	No	No	No	Yes
Nonce Validation	N/A	No	No	No	Yes
GPS Evidence	No	No	Yes	No	Yes
BLE Evidence	No	No	No	Yes	Yes
JWT-Based Auth.	No	No	No	No	Yes
Administrative Governance	Limited	No	Limited	Limited	Yes
Cloud Deployment	Rarely	Rarely	Sometimes	Rarely	Yes

2.14 Review of Related Research Papers

To situate HKS ATTENDANCE within the published research on attendance automation, Table 2.4 summarises twelve representative papers and standards spanning BLE proximity systems, GPS-based systems, multi-factor and hybrid verification schemes, and the underlying wireless and cryptographic primitives the project draws on. Full bibliographic details for each entry, together with the remaining references consulted during the project, appear in the References section at the end of this thesis.

Across this body of work, very few systems combine more than one verification channel, almost none bind a student account to a specific device, and fewer still provide a structured path for a student to contest a rejected record or allow faculty to inspect the underlying evidence before a decision is finalised. Table 2.5 expresses this comparison as a feature-coverage matrix.

2.15 Observations Arising from the Research Gap Analysis

Reading Tables 2.2 through 2.5 together yields five specific observations that directly shaped the requirements presented in Chapter 3:

1. **Observation 1** – No reviewed system combines BLE and GPS evidence in a sin-

Table 2.4: Summary of Reviewed Research Papers

#	Focus Area	Methodology	Reported Advantage	Reported Limitation
1	BLE-based attendance management [1]	Fixed BLE beacons paired with a mobile scanning application	Removes manual roll-call effort; low beacon power draw	No device-binding or replay protection described
2	Smart attendance using BLE [2]	RSSI-threshold proximity detection	Improves on RFID by removing the bearer-card problem	Signal strength alone reported as noisy and position-sensitive
3	GPS-based attendance for mobile devices [3]	Geofence radius check against a single registered campus location	No additional hardware required	Cannot distinguish adjacent rooms; vulnerable to location spoofing
4	Multi-factor secure attendance [4]	Combines a password factor with a location factor	Reduces simple credential sharing	Does not address device binding or proximity verification
5	Survey of wireless attendance technologies [5]	Comparative literature survey	Identifies that no single wireless technology is sufficient alone	Survey-level; does not implement a combined system
6	Bluetooth Core Specification [6]	Protocol-level specification of BLE advertising behaviour	Defines the low-power advertising model the project's beacons rely on	Specification only; no application-layer security guidance
7	ESP32 technical reference [7, 8]	Hardware and firmware reference for ESP32-class microcontrollers	Establishes feasibility of low-cost, long-running beacon hardware	Does not address attendance-specific payload design
8	Bluetooth/Wi-Fi interference study [9]	Simulation of 802.11 and Bluetooth coexistence	Quantifies expected interference in shared 2.4 GHz environments	Does not address spoofing or replay threats
9	Mobile location services documentation [10, 11]	Platform-level GPS and fused-location APIs	Provides accuracy and freshness metadata usable for evidence quality checks	Documentation rather than a verification scheme
10	HMAC keyed-hashing specification [16]	Cryptographic primitive for message authentication	Provides the signature mechanism used to authenticate beacon payloads	Generic primitive; requires application-specific key and nonce management
11	JSON Web Token specification [17]	Token-based authentication standard	Provides a stateless, verifiable session credential	Does not address device binding or proximity verification
12	OWASP API Security Top 10 [15]	Catalogue of common API security weaknesses	Used to validate the security posture of the REST interface	General-purpose; not specific to attendance systems

Table 2.5: Research Gap Matrix

System / Approach	BLE	GPS	Device Binding	Faculty Review	Claims Resolution
Manual Register	No	No	No	No	No
RFID Card Reader	No	No	No	No	No
Scannable Code	No	No	No	No	No
GPS-Only Mobile App	No	Yes	No	No	No
BLE-Only Mobile App [1, 2]	Yes	No	No	No	No
Multi-Factor Scheme [4]	No	Limited	No	No	No
HKS ATTENDANCE (this work)	Yes	Yes	Yes	Yes	Yes

gle, jointly evaluated decision; where both are present, they are typically used as alternatives rather than as corroborating signals.

2. **Observation 2** – Device binding is essentially absent from the literature reviewed, despite credential and card sharing being the most frequently cited weakness of existing systems.
3. **Observation 3** – Faculty governance, in the sense of a human reviewer inspecting raw evidence before a decision is finalised, appears in very few systems; most either accept or reject automatically.
4. **Observation 4** – A structured claims or appeals workflow is rarely implemented, leaving students with no formal recourse when an automated decision is incorrect.
5. **Observation 5** – Replay protection for wireless evidence (BLE or otherwise) is largely absent from the attendance-specific literature, even though it is a well-understood requirement in the broader wireless-security literature.

Each of these five observations maps directly onto one of the six design choices listed in Section 2.16, and onto the functional requirements set out in Chapter 3.

2.16 Positioning of HKS ATTENDANCE

HKS ATTENDANCE responds to the gaps identified above through six design choices acting together rather than in isolation:

- Attendance decisions are derived jointly from BLE proximity evidence and GPS geofence evidence, rather than from either alone.
- Every attendance submission is tied to a single device registered against the submitting student’s account.
- Beacon broadcasts are protected with single-use nonces, freshness checks, and cryptographic signatures to resist replay and spoofing.
- Submissions that the automated decision engine cannot confidently accept or reject are routed to the responsible faculty member for evidence-based review.
- Administrative tooling manages the full academic hierarchy – users, courses, enrolments, classrooms, and trusted beacons – as a first-class part of the platform.

- Students may formally contest a rejected attendance record, with the outcome and reasoning permanently logged.

2.17 Chapter Summary

This chapter reviewed the principal technologies used or proposed for automated attendance – registers, RFID, scannable codes, GPS, biometrics, BLE, facial recognition, and combinations thereof – together with a representative set of published research papers and standards addressing BLE- and GPS-based verification, and positioned HKS ATTENDANCE against a representative set of commercial attendance products. Five specific observations were drawn from the resulting gap analysis, and a consistent pattern emerged: systems built around a single evidence channel inherit that channel’s specific weakness, while hybrid designs trade implementation complexity for stronger resistance to manipulation. The next chapter translates these observations into a concrete set of functional and non-functional requirements for HKS ATTENDANCE.

Chapter 3

System Analysis

3.1 Introduction

Before development began, the project required a clear analysis of what existing attendance practices fail to deliver, which of those failures a new system could realistically address, and what a workable solution would need to do in concrete, testable terms. This chapter presents that analysis: the shortcomings of current practice, the requirements they imply, a feasibility assessment across technical, economic, operational, and scheduling dimensions, and the constraints and assumptions under which the system was designed and built.

3.2 Shortcomings of Current Practice

3.2.1 Impersonation

The most persistent problem in attendance recording is one student marking presence on behalf of an absent classmate. Manual registers offer no defence against this, and many digitised systems do little better, since a shared credential, a borrowed card, or a forwarded screenshot reproduces the same failure in a different form.

3.2.2 Single-Channel Verification

Most automated systems reviewed in Chapter 2 base the entire attendance decision on one piece of evidence. Once that channel is compromised – a card is lent, a code is forwarded, a location is spoofed – there is no remaining basis for the decision.

3.2.3 Coarse Location Granularity

Systems that rely on GPS alone frequently confirm only that a device is somewhere on or near a campus, not that it is in the specific room where a session is taking place. A student outside a lecture hall, or in a neighbouring building under favourable signal conditions, can sometimes register a presence that would not withstand closer scrutiny.

3.2.4 Weak Auditability

When an attendance dispute arises, institutions are frequently unable to reconstruct why a particular decision was made, what evidence supported it, or who reviewed it. Without this trail, disputes tend to be resolved informally rather than on the basis of recorded fact.

3.2.5 Absence of Governance Workflow

Many systems treat attendance as a write-once operation: a submission is accepted or rejected, and that is the end of the matter. There is no path for a faculty member to intervene in an ambiguous case, and no path for a student to formally contest an outcome believed to be wrong.

3.2.6 Security Weaknesses

Common weaknesses observed across the systems reviewed include vulnerability to replayed verification data, ease of sharing credentials or devices between students, the possibility of an unauthorised beacon or reader masquerading as a trusted one, and a lack of enforcement around session validity, allowing attendance to be submitted after a class has ended.

3.3 The Case for a New System

Taken together, these shortcomings point toward a system that verifies physical presence through more than one channel, ties submissions to an authenticated and individually identifiable device, gives faculty visibility into the evidence behind ambiguous decisions, gives students a formal way to contest outcomes, and records every governance action permanently. These requirements directly shaped the design of HKS ATTENDANCE.

3.4 Proposed System Overview

HKS ATTENDANCE bases its attendance decisions on the combination of BLE proximity evidence and GPS geofence evidence, layered with device-binding enforcement, faculty review of ambiguous submissions, administrative governance of the academic hierarchy and beacon infrastructure, and a formal claims process for disputed outcomes.

3.5 Architecture at a Glance

The platform follows a layered design: a Flutter mobile client at the top, communicating over a REST interface with a FastAPI backend, which enforces business rules and security checks before persisting state to a PostgreSQL database. The technology choices are summarised in Table 3.1.

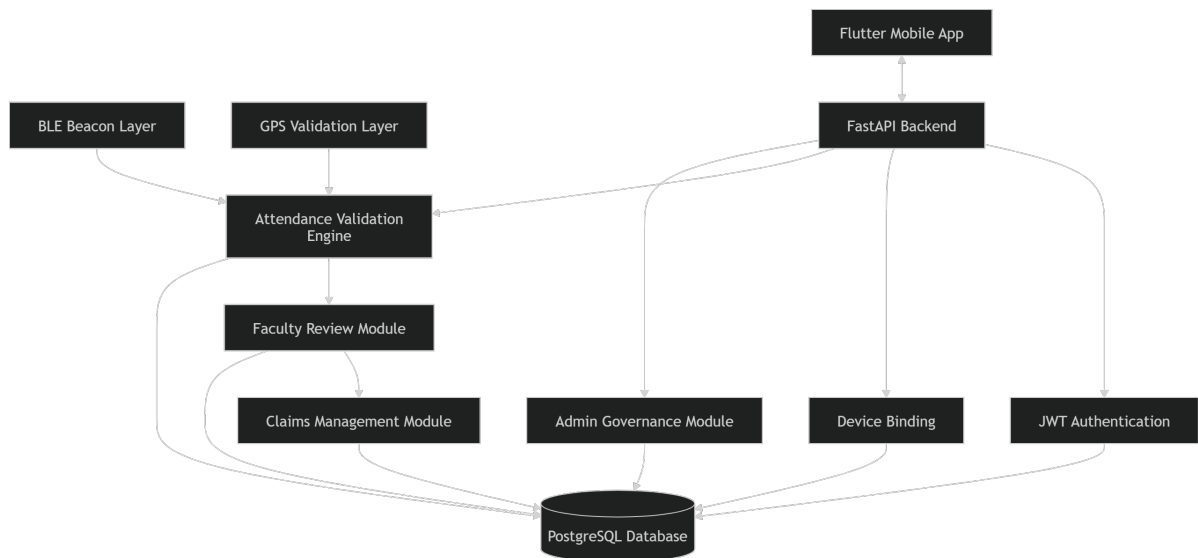


Figure 3.1: Overall System Architecture

Diagram redrawn for this submission from the layer descriptions given throughout Chapters 3–6.

3.6 Stakeholders and Responsibilities

3.6.1 Student

- Authenticate and discover sessions for enrolled courses.

Table 3.1: Core Technology Choices

Layer	Technology
Mobile Application	Flutter
Backend API	FastAPI
Database	PostgreSQL
BLE Infrastructure	ESP32 microcontrollers
Authentication	JSON Web Tokens (JWT)

- Collect and submit BLE and GPS evidence to register attendance.
- View attendance history and profile information.
- Submit claims against rejected attendance records.

3.6.2 Faculty

- Start, monitor, and close attendance sessions.
- Review flagged attendance and inspect supporting evidence.
- Confirm or reject ambiguous attendance records.
- Review and resolve student claims.

3.6.3 Administrator

- Manage student and faculty accounts.
- Manage courses, faculty-course assignments, and enrolments.
- Manage classrooms and the trusted-beacon registry.
- Govern device bindings, including unbinding compromised or lost devices.

Table 3.2: Summary of Functional Requirements

ID	Requirement
FR-1	Authenticate users and issue session-scoped JWT tokens
FR-2	Bind a student account to exactly one active device
FR-3	Surface only active sessions for a student's enrolled courses
FR-4	Accept BLE and GPS evidence as part of attendance submission
FR-5	Validate BLE evidence against the trusted-beacon registry
FR-6	Validate GPS evidence against the relevant classroom geofence
FR-7	Present flagged attendance records for faculty review
FR-8	Allow faculty to confirm or reject flagged records
FR-9	Allow students to submit and track claims; allow faculty to resolve them
FR-10	Provide administrative CRUD operations over the academic hierarchy

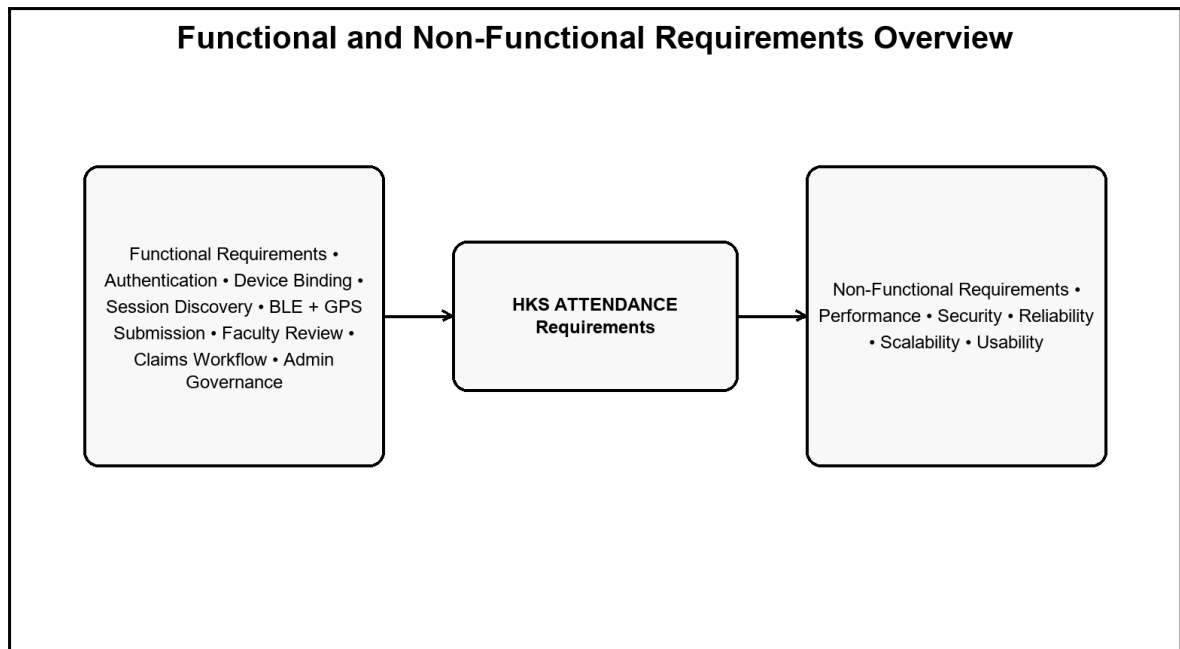


Figure 3.2: Functional and Non-Functional Requirements Overview

3.7 Functional Requirements

3.8 Non-Functional Requirements

3.8.1 Performance

Attendance submission, including evidence validation, was targeted to complete within approximately three seconds under normal operating conditions.

3.8.2 Security

The system must enforce authentication, role-based authorisation, device verification, and protection against replayed evidence.

3.8.3 Reliability

Attendance operations must complete without leaving the database in an inconsistent state, even under concurrent submissions.

3.8.4 Scalability

The schema and architecture must accommodate institutional growth in the number of students, courses, sessions, and historical attendance records without requiring redesign.

3.8.5 Usability

Interfaces for students, faculty, and administrators must remain intuitive enough to require minimal training.

3.9 Feasibility Study

3.9.1 Technical Feasibility

Every technology selected for the project – Flutter, FastAPI, PostgreSQL, JWT, and ESP32-based BLE firmware – is mature, well-documented, and widely supported, making the project technically feasible within the available development timeframe.

3.9.2 Economic Feasibility

Beyond the smartphones and network connectivity that students and staff already possess, the only additional hardware required is a small number of inexpensive ESP32 boards per classroom, keeping the marginal deployment cost low relative to card-reader or biometric alternatives.

3.9.3 Operational Feasibility

Each stakeholder group interacts with the system through a role-specific interface – a student app, a faculty dashboard, and an administrative console – none of which demands specialised training beyond a short orientation.

3.9.4 Schedule Feasibility

The project was structured into eight sequential phases – academic-model design, session discovery, the evidence layer, the student-facing product, faculty governance, administrative governance, the claims subsystem, and final validation – each completed within the academic timeline allotted to the project.

3.9.5 Consolidated Feasibility Summary

Table 3.3 consolidates the feasibility assessment across all dimensions considered for the project, including legal and risk considerations not discussed as separate subsections above.

3.10 Constraints

- Attendance submission requires an active internet connection.
- BLE evidence collection requires properly installed and registered classroom beacons.
- GPS evidence collection requires location services to be enabled on the student's device.
- Every student is assumed to possess a compatible smartphone.

3.11 Assumptions

- Classroom coordinates and geofence radii are configured accurately by administrators.
- Beacons are installed in the classroom they are registered against and not relocated without an administrative update.
- Faculty and administrative accounts are used only by their authorised holders.

3.12 Anticipated Benefits

- Stronger resistance to impersonation through device binding and dual-channel verification.
- Improved auditability through comprehensive evidence and decision logging.
- Shared governance between automated checks and human faculty review.
- A structured, transparent path for resolving attendance disputes.

3.13 Chapter Summary

This chapter analysed the shortcomings of existing attendance practice, translated them into functional and non-functional requirements, and confirmed the technical, economic, operational, and scheduling feasibility of building HKS ATTENDANCE within the available project timeline. The next chapter presents the resulting system design in detail.

Table 3.3: Consolidated Feasibility Summary

Dimension	Assessment	Status
Technical	All selected technologies are mature, well-documented, and within the team's existing skill set.	Feasible
Economic	Marginal hardware cost is limited to inexpensive ESP32 beacons, with no licensing cost for the core stack.	Feasible
Operational	Role-specific interfaces require minimal training for students, faculty, and administrators.	Feasible
Legal	No biometric data is collected; location data is used only transiently for geofence evaluation.	Feasible, subject to institutional policy
Schedule	The eight-phase implementation plan fits within the academic project timeline.	Feasible
Risk	Main risks are beacon misconfiguration and indoor GPS variability, both mitigated by faculty review and validation controls.	Manageable

Chapter 4

System Design

4.1 Introduction

This chapter translates the requirements established in Chapter 3 into a concrete architecture: the layering of the system, the technologies bound to each layer, the major functional modules, and the way evidence flows from a student’s phone through to a final, faculty-reviewable attendance decision.

4.2 Design Principles

Before describing the architecture itself, it is useful to state the design principles applied consistently across every layer and module of HKS ATTENDANCE:

- **Security by Design** – authentication, device binding, and evidence validation are treated as core requirements present from the first design iteration, not as features added after a working prototype existed.
- **Evidence Fusion** – no single signal is ever trusted in isolation; every attendance decision draws on at least two independent evidence sources, as detailed in Section 4.5.
- **Separation of Concerns** – presentation, business logic, security, and persistence are kept in distinct layers (Section 4.3), so that a change to one (for example, swapping the BLE library used on the client) does not require changes to the others.
- **Least Privilege** – each role (student, faculty, administrator) is granted only the permissions its tasks require; a faculty member cannot, for instance, act on a course they do not own.
- **Role-Based Access** – every protected endpoint checks the caller’s role and, where relevant, their ownership of the specific resource being acted upon, before performing the requested action.

- **Scalability** – the schema and architecture are normalised around courses, sessions, and classrooms rather than hard-coded institutional specifics, so that additional courses, classrooms, or even departments can be added without structural change.
- **Modularity** – the six functional modules introduced in Section 4.6 are designed to evolve largely independently of one another.

4.3 Layered Architecture

The system is organised into five layers, each responsible for a distinct concern: the Flutter mobile client at the presentation layer; a FastAPI-based REST interface; a business-services layer implementing attendance, claims, faculty, and administrative logic; a security layer enforcing authentication, BLE and GPS validation, and device checks; and a PostgreSQL persistence layer at the base. Requests flow downward from the mobile client through each layer in turn, with each layer validating only the concerns it owns before handing off to the next.

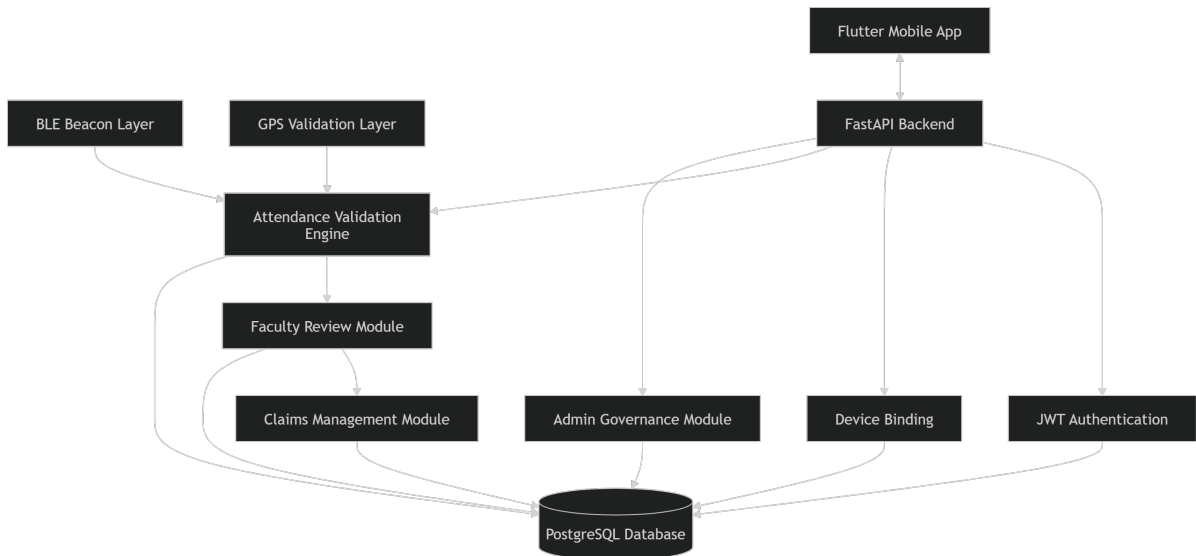


Figure 4.1: Layered Architecture Diagram

Table 4.1: Implementation Technologies by Layer

Layer	Technology
Mobile Application	Flutter / Dart
Backend API	FastAPI (Python)
Database	PostgreSQL
ORM	SQLAlchemy
Migrations	Alembic
Authentication	JWT
BLE Infrastructure	ESP32
Transport	REST over HTTPS

4.4 Technology Choices Revisited

4.5 Evidence-Fusion Philosophy

The architectural premise underlying the system is that attendance should not be decided on the strength of a single signal. Every submission is evaluated against two independent evidence sources – BLE proximity and GPS location – combined with a device-identity check, before a decision is reached. The attendance engine itself does not autonomously issue a final rejected outcome. Instead, it either confirms a submission when the collected evidence is sufficiently trustworthy, flags the submission for faculty review when the evidence is weak, inconsistent, or suspicious, or blocks the submission entirely when no BLE evidence is captured. This design keeps final rejection as a faculty governance action rather than an automatic system decision.

4.6 Major Functional Modules

The system is organised into six cooperating modules:

- **Student Module** – attendance submission and history.
- **Faculty Module** – session governance and attendance review.
- **Admin Module** – academic and infrastructure governance.
- **BLE Module** – beacon discovery and proximity validation.
- **GPS Module** – location capture and geofence validation.
- **Claims Module** – dispute submission and resolution.

4.7 Student Module

The student-facing module supports login, discovery of active sessions scoped to the student’s enrolments, evidence collection and attendance submission, viewing of attendance history, claim submission, and profile management. The attendance flow proceeds as: authenticate, discover an eligible session, collect BLE evidence, collect GPS evidence, submit, and receive an immediate system response. The automated outcome is either **CONFIRMED** or **FLAGGED for faculty review**. If the submission fails a mandatory precondition — such as missing BLE evidence, invalid session state, duplicate submission, stale or replayed BLE payload, or device-binding failure — the request is blocked and no attendance record is created.

4.8 Faculty Module

Faculty members start and close attendance sessions, monitor submissions as they arrive, inspect the underlying BLE and GPS evidence for any flagged record, and confirm or reject those records with a recorded reason. A faculty dashboard surfaces active sessions, pending reviews, and pending claims at a glance.

4.9 Administrative Module

Administrators manage the full academic hierarchy – students, faculty, courses, faculty-course assignments, enrolments, classrooms – together with the trusted-beacon registry and device-binding lifecycle (including unbinding a device when a student loses or replaces their phone).

4.10 BLE Module

Each classroom beacon, built on an ESP32 microcontroller, periodically broadcasts a small payload containing a beacon identifier, a freshly generated nonce, a timestamp, and an HMAC signature computed over the payload using a secret known only to the backend and that specific beacon. The mobile application scans for and reads this payload, and the backend independently recomputes the signature to confirm the broadcast is genuine before accepting it as evidence.

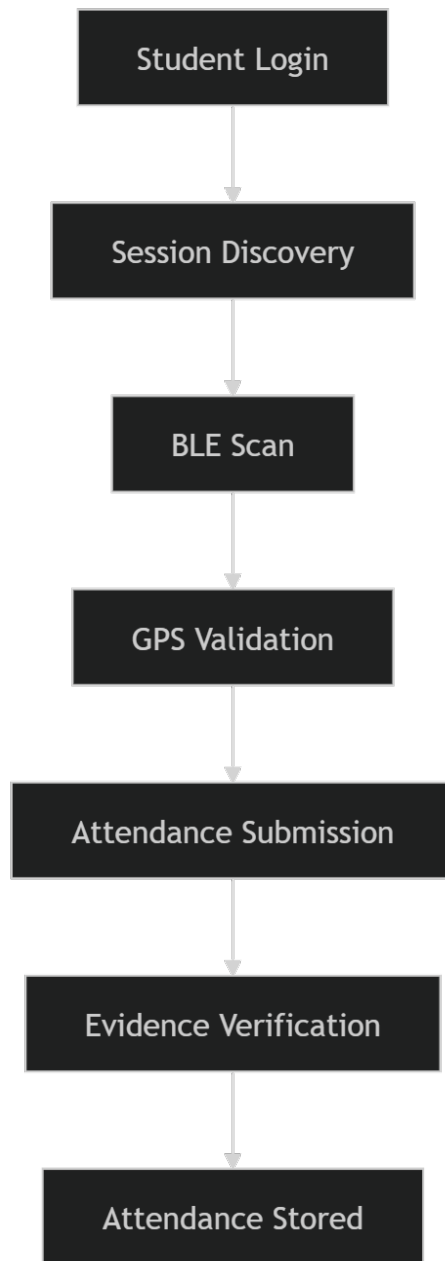


Figure 4.2: Student Attendance Workflow

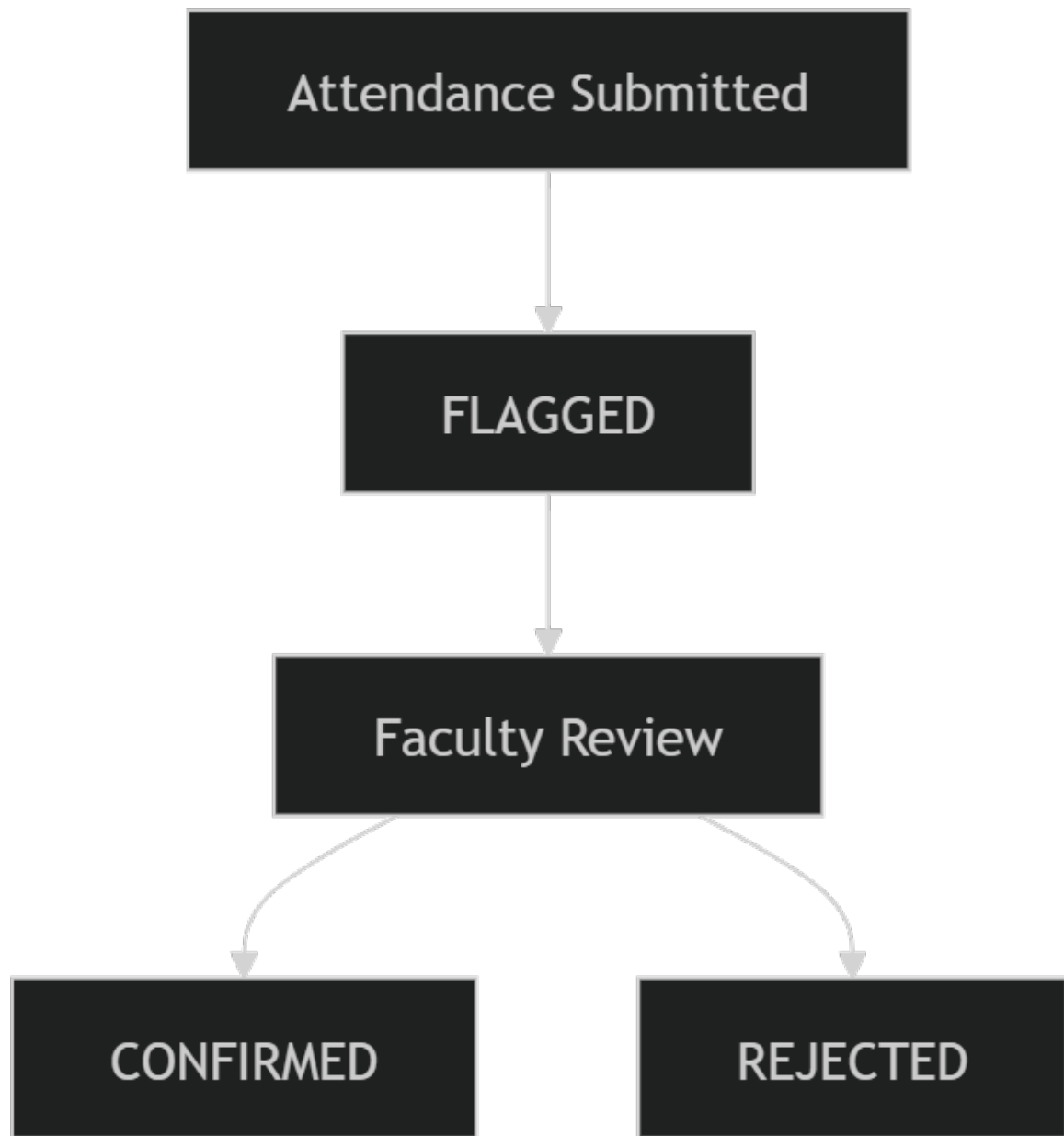


Figure 4.3: Faculty Review Workflow

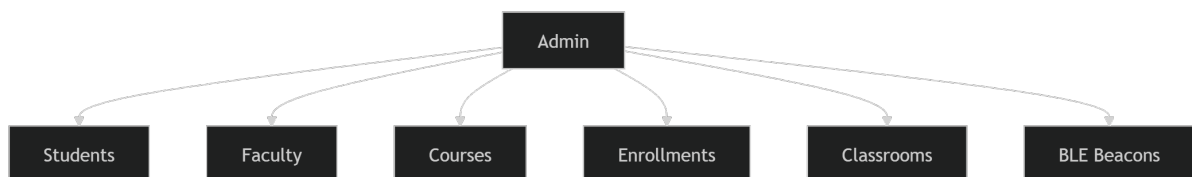


Figure 4.4: Admin Governance Workflow

Example BLE Beacon Payload:

```
{  
  "beacon_id": "ESP32-A1",  
  "nonce": "xyz123",  
  "timestamp": 1750000000,  
  "signature": "..."  
}
```

4.11 GPS Module

The mobile application captures the device's latitude, longitude, and reported accuracy at the moment of submission. The backend computes the distance between the reported coordinates and the classroom's registered location and checks that distance against the classroom's configured geofence radius, producing one of three outcomes: inside the geofence, outside the geofence, or accuracy too low to evaluate reliably.

4.12 Evidence-Fusion Decision Table

The matrix in Table 4.2 represents the automated outcome produced by the attendance engine only. A record marked as FLAGGED is not rejected automatically; instead, it is routed to the responsible faculty member for evidence-based review. If no BLE evidence is captured, the submission is blocked and no attendance record is created. Final rejection, where applicable, is therefore a faculty governance decision rather than an automatic system outcome.

4.13 Sequence Diagram: Attendance Submission

Figure 4.6 traces a single attendance submission as a sequence of messages between the student, the Flutter client, the backend API, the BLE and GPS validators, the decision engine, and the database. This view makes explicit the order in which evidence is collected and validated, and clarifies that the decision engine only acts once both evidence channels have returned a result.

Table 4.2: Attendance Decision Matrix

BLE Result	GPS Result	Decision
Valid	Valid	CONFIRMED
Valid	Invalid / Outside Geofence / Low Accuracy	FLAGGED for faculty review
Invalid / Weak / Suspicious	Valid	FLAGGED for faculty review
Invalid / Weak / Suspicious	Invalid / Outside Geofence / Low Accuracy	FLAGGED for faculty review
No BLE evidence captured	Any GPS result	Submission blocked; no attendance record is created

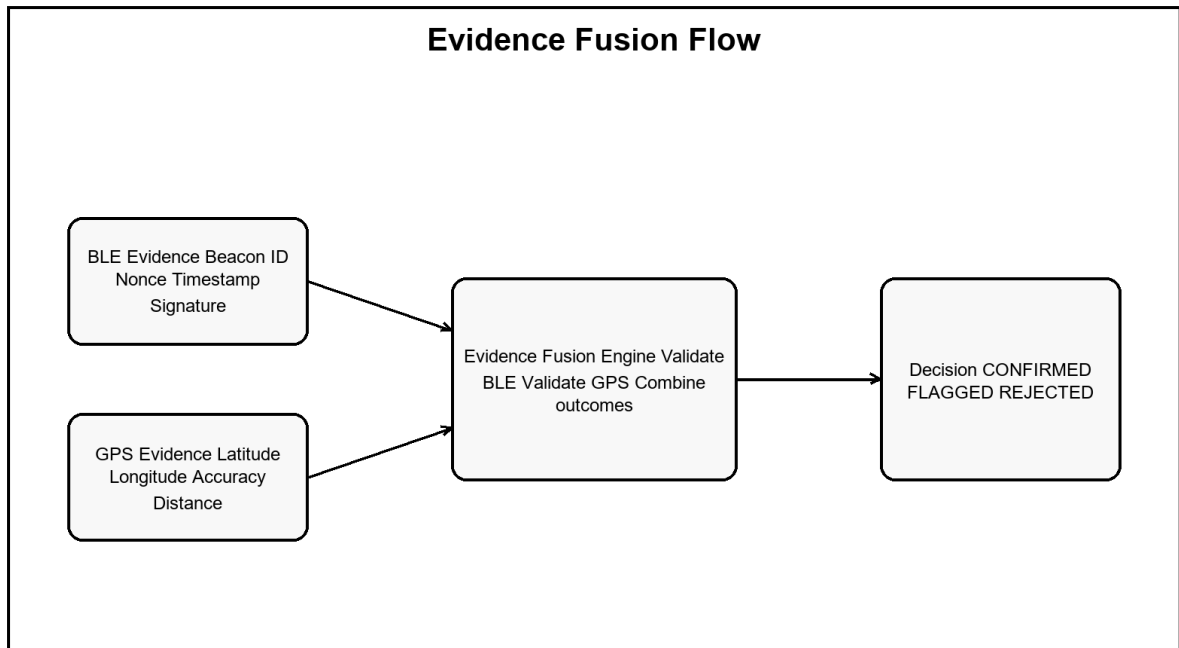


Figure 4.5: Evidence Fusion Flow

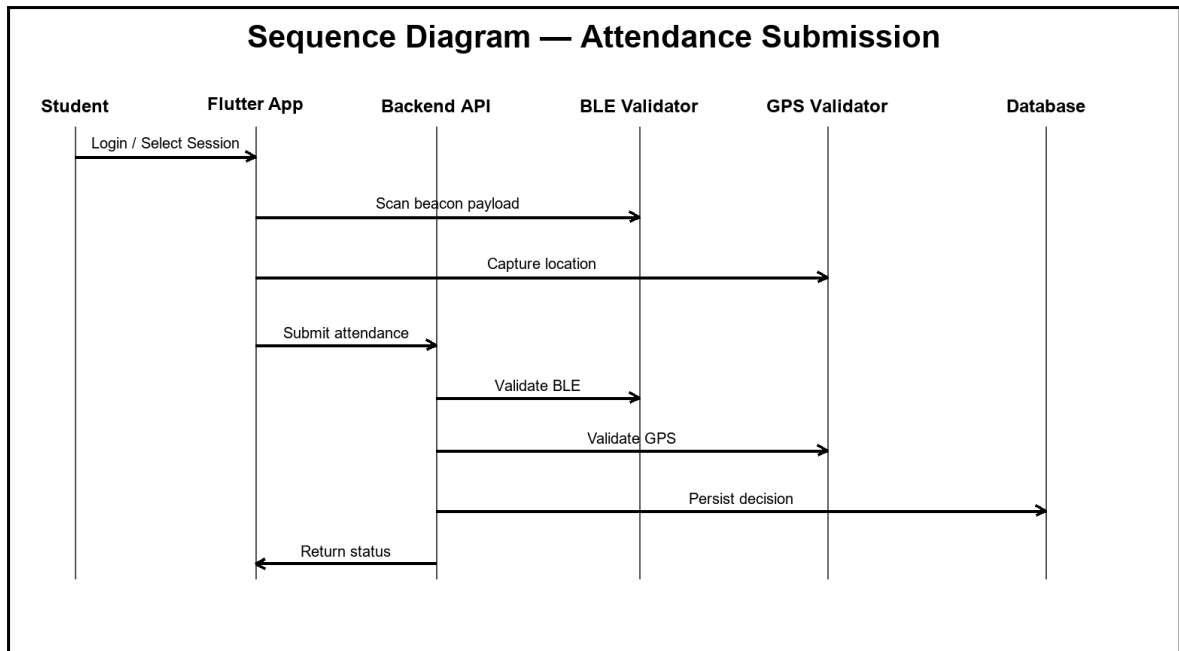


Figure 4.6: Sequence Diagram – Attendance Submission

4.14 Security Architecture

To make the security reasoning behind the layered architecture of Section 4.3 explicit, Figure 4.7 traces a single request as it passes through every security checkpoint between the student’s device and the persisted record, restated below as a checkpoint pipeline.

```

Student Device
  |
  v
JWT Authentication (identity + role)
  |
  v
Backend API (FastAPI)
  |
  v
BLE Validation (nonce, freshness, HMAC signature)
  |
  v
GPS Validation (geofence distance, accuracy)
  |
  v
Evidence Fusion / Decision Engine
  
```

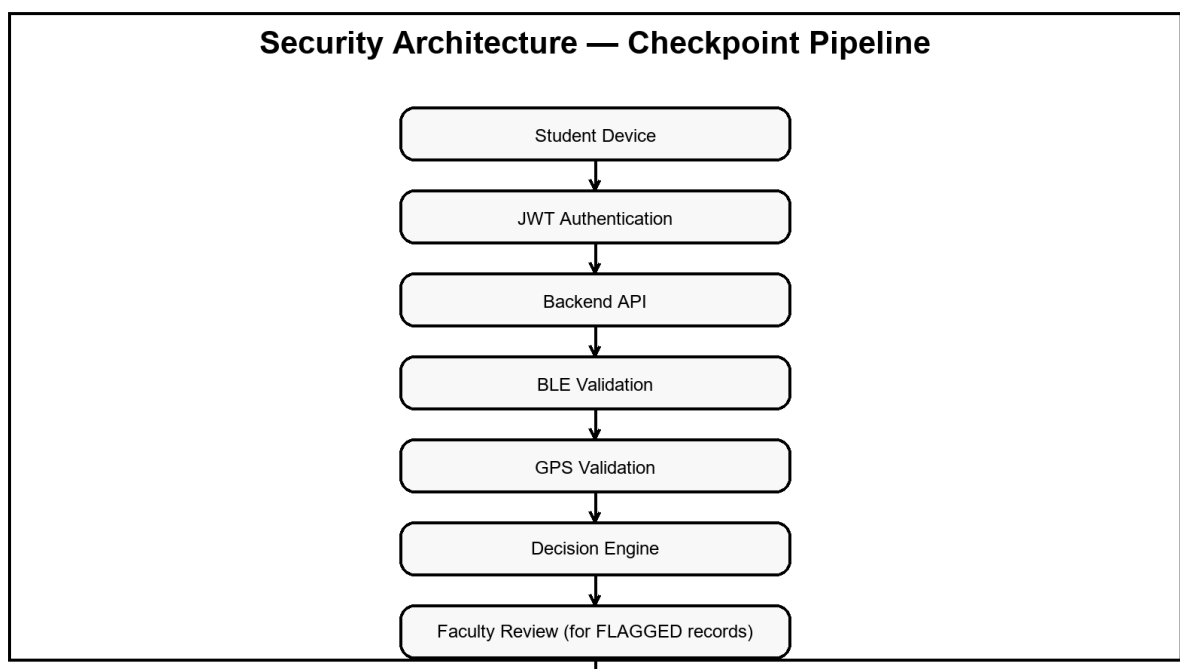
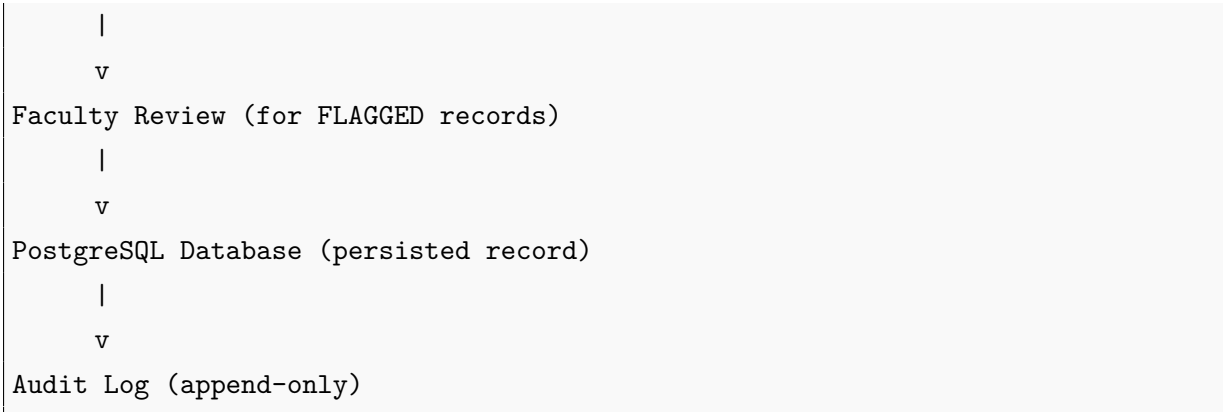


Figure 4.7: Security Architecture – Checkpoint Pipeline

Each checkpoint in this pipeline is independently capable of rejecting a request, so that no single compromised layer is sufficient to produce an unwarranted CONFIRMED outcome; this is the architectural counterpart of the threat model presented in Table 6.3.

4.15 Data Flow Diagram (Level 0)

Figure 4.8 presents a context-level (Level 0) data-flow diagram for the system, showing the three external actors – Student, Faculty, and Administrator – and their principal data exchanges with the HKS ATTENDANCE platform as a single process.

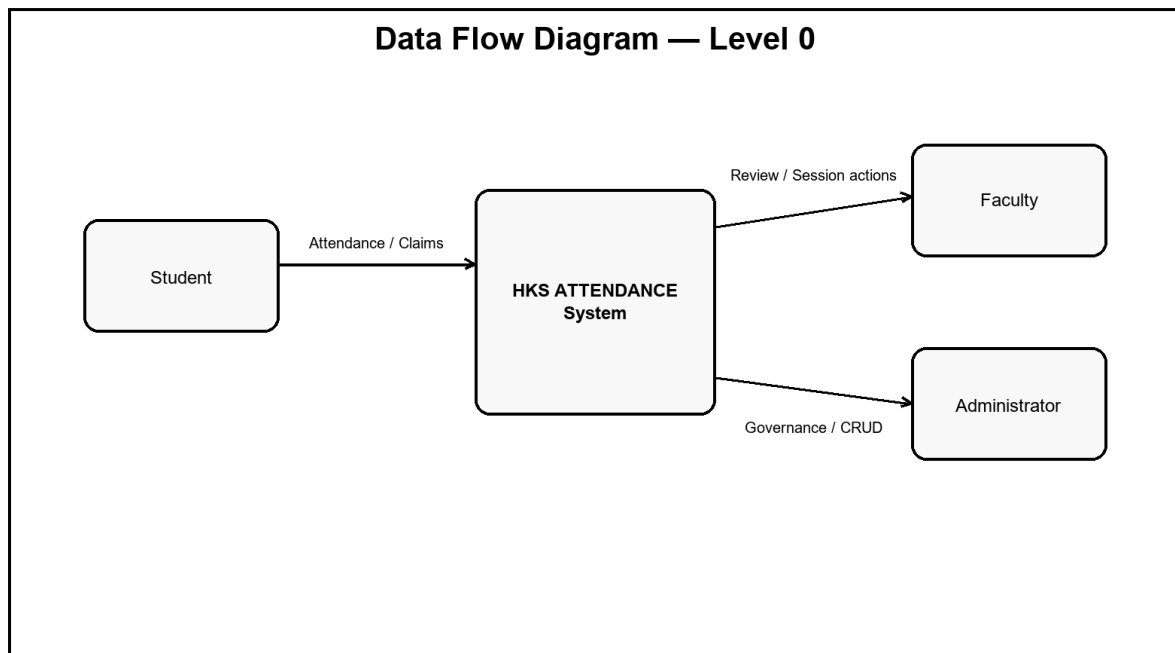


Figure 4.8: Data Flow Diagram – Level 0 (Context Diagram)

4.16 Component and Package Diagrams

Figure 4.9 shows the major software components within the backend (routers, services, security, and data-access layers) and their dependencies, while Figure 4.10 shows the corresponding package-level organisation of the Flutter mobile client described in Section 6.3.

4.17 Authentication Design

Login credentials are validated against the stored, hashed password, and a JWT access token is issued on success. The token is attached to all subsequent requests and is required for attendance submission, claims, and any faculty or administrative action.

4.18 Device-Binding Design

A student’s first successful login registers the device they are using as their bound device. Subsequent attendance submissions are checked against this binding; a request originating from an unrecognised device is rejected outright. Should a student need to switch devices, an administrator must first unbind the old device, after which the student’s next successful

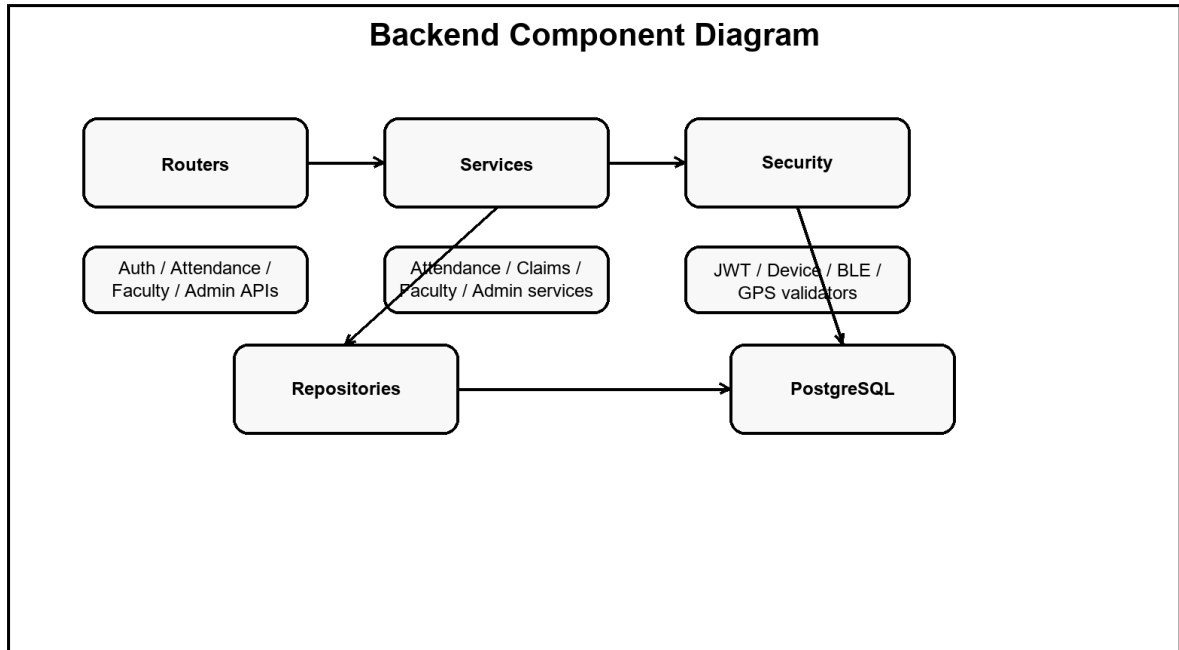


Figure 4.9: Backend Component Diagram

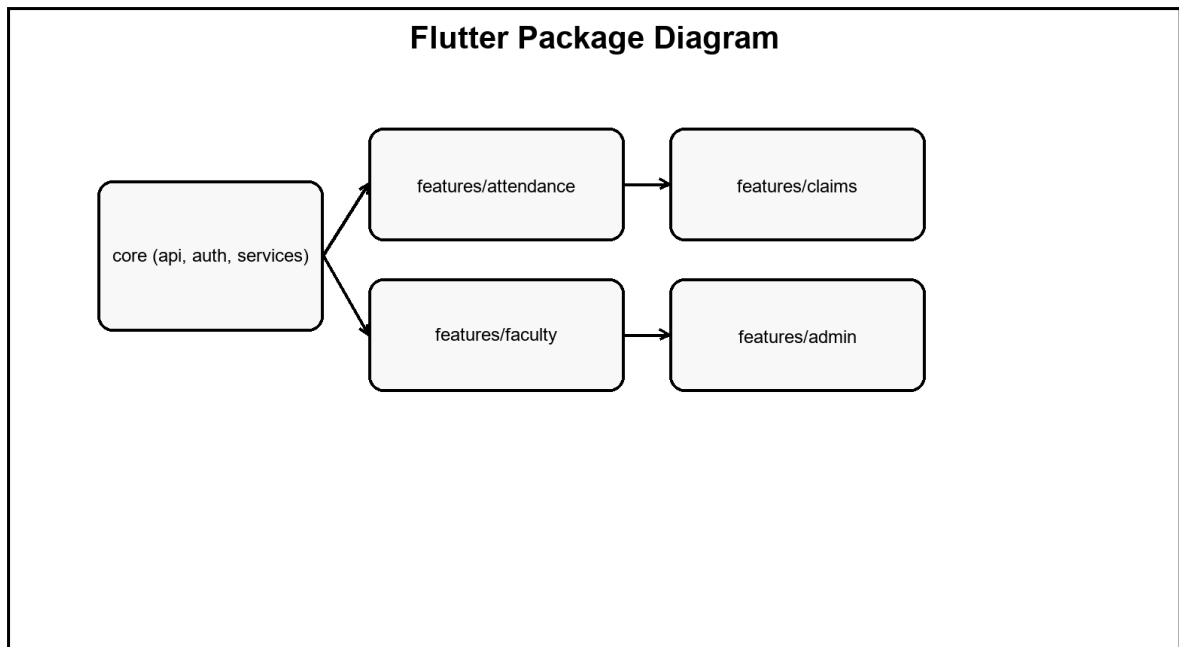


Figure 4.10: Mobile Application Package Diagram

login registers the replacement.

4.19 Claims Module Design

Only attendance records in the REJECTED state may be claimed, and only one claim may be opened per attendance record. A submitted claim moves through PENDING, and is resolved by the faculty member who owns the relevant course into either APPROVED – which moves the underlying attendance record to CONFIRMED – or REJECTED, which leaves the original outcome unchanged. Every resolution is written to the audit log.

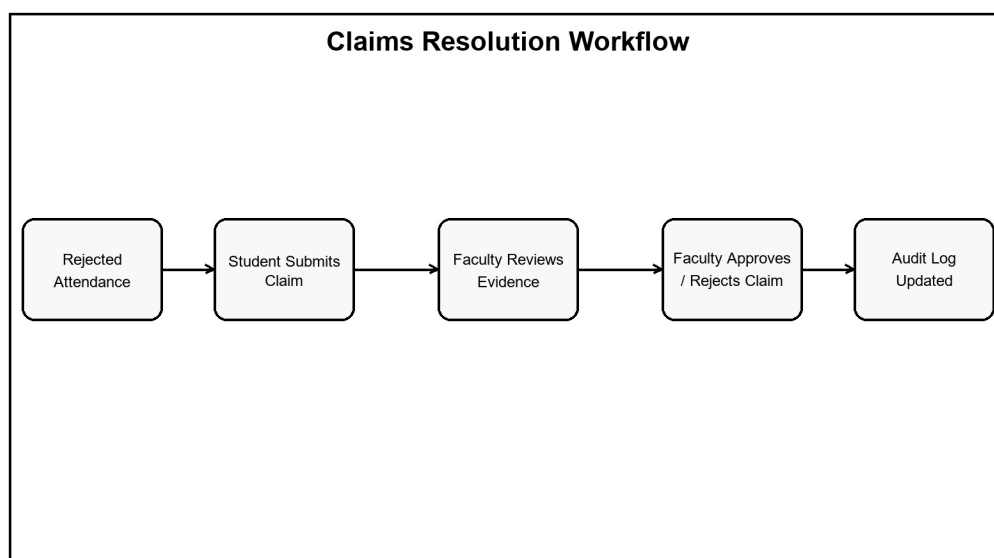


Figure 4.11: Claims Resolution Workflow

4.20 Audit-Logging Design

Every governance-relevant action – confirming or rejecting attendance, approving or rejecting a claim, unbinding a device, closing a session – produces a permanent log entry recording the actor, the action, the before-and-after state, and a timestamp.

4.21 Session-Lifecycle Design

Attendance sessions move through three states: ACTIVE, CLOSED (closed manually by faculty), and EXPIRED (closed automatically once the configured duration elapses).

Attendance may only be submitted while a session is ACTIVE.

4.22 Attendance Status State Diagram

Independently of session lifecycle, each individual attendance attempt carries its own status, which moves through a small set of well-defined states from creation to final resolution, including the claims path for records that are initially rejected. Figure 4.12 shows this state diagram.

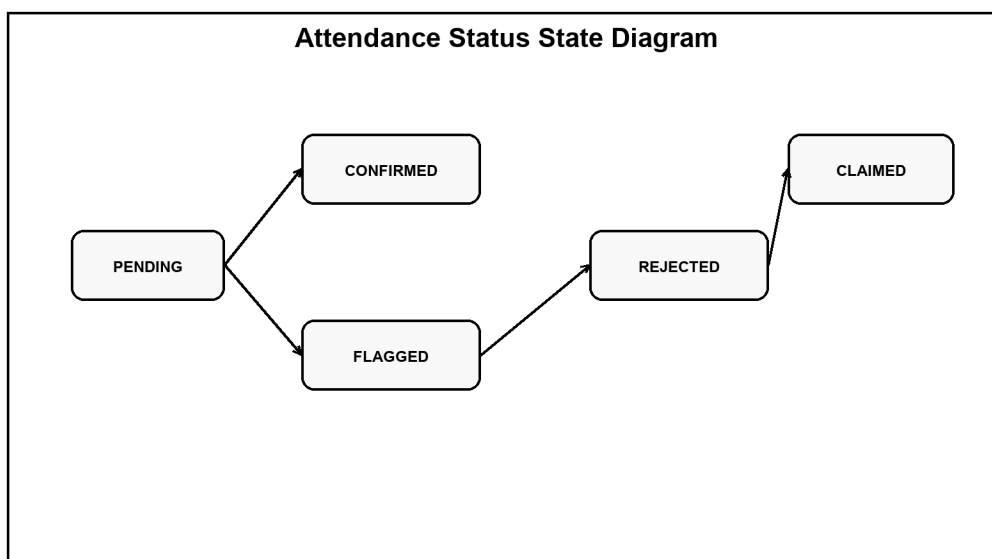


Figure 4.12: Attendance Status State Diagram

4.23 Use-Case Summary

Table 4.3: Representative Use Cases by Role

Role	Representative Use Cases
Student	Login; discover sessions; submit attendance; view history; submit claims
Faculty	Start/close sessions; review attendance; resolve claims
Administrator	Manage users, courses, enrolments, classrooms, beacons, and devices

4.24 Design Advantages

- **Security:** decisions rest on multiple independent validation layers.

- **Governance:** faculty and administrators retain meaningful oversight.
- **Auditability:** every consequential action is logged.
- **Modularity:** the six functional modules can evolve largely independently.
- **Extensibility:** the architecture leaves room for additional evidence sources in the future.

4.25 Chapter Summary

This chapter presented the layered architecture, the major functional modules, the evidence-fusion logic, the security-checkpoint pipeline, and the context-, component-, and package-level diagrams at the heart of HKS ATTENDANCE. The next chapter describes the database design that underpins this architecture.

Chapter 5

Database Design

5.1 Introduction

Every attendance decision, governance action, and dispute resolution in HKS ATTENDANCE ultimately rests on data persisted in PostgreSQL. This chapter describes the relational schema underpinning the platform, the reasoning behind its major entities, and the integrity constraints that keep the academic and security records consistent. PostgreSQL was chosen for its ACID guarantees, mature support for foreign-key enforcement, and proven reliability under the kind of transactional load an attendance system generates.

5.2 Design Objectives

- Preserve attendance records as immutable, accurate history once written.
- Support faculty review and administrative governance directly through the schema.
- Store BLE evidence, GPS evidence, device bindings, and audit entries durably.
- Scale comfortably as the number of students, courses, sessions, and historical records grows.
- Preserve enough history to support after-the-fact dispute investigation.

5.3 Database Design Philosophy

Before describing individual tables, it is useful to state the design principles applied consistently across the schema:

- **Normalisation** – the schema is normalised to third normal form (Section 5.4), so that each fact is stored exactly once and update anomalies are avoided.

- **Primary keys** – every table is keyed by a surrogate integer or UUID primary key, kept independent of any business attribute that might change over time.
- **Foreign keys** – every cross-table reference (a session referencing a course, an attempt referencing a session, and so on) is enforced as a foreign-key constraint at the database level, not merely checked in application code.
- **Indexes** – foreign-key columns and columns used in frequent lookups (for example, a student’s active sessions, or a beacon’s nonce history) are indexed to keep query latency low as the volume of historical data grows.
- **Constraints** – uniqueness constraints (for example, one device per student, one open claim per attendance attempt) are enforced in the schema itself, so that the invariant holds even if application-level validation is bypassed or contains a bug.
- **Transactions** – multi-step writes (for example, approving a claim and updating the underlying attendance record) are wrapped in a single database transaction, so that a failure partway through cannot leave the two records in an inconsistent state.
- **Referential integrity** – the database, not the application, is the final authority on whether a reference between two rows is valid; this is treated as a security boundary as much as a correctness one (Section 5.9).

5.4 Normalisation Approach

The schema was normalised to third normal form. Atomic fields (for example, separate first- and last-name columns rather than a single combined name field) satisfy first normal form; every non-key attribute depends on the whole of its table’s primary key, satisfying second normal form; and course information is stored once in a Courses table rather than repeated on every attendance record, satisfying third normal form and avoiding the update anomalies that denormalised duplication would otherwise introduce.

5.5 Entity Overview

The principal entities are Users, Devices, Courses, FacultyCourses, Enrollments, Classrooms, TrustedBLEBeacons, BeaconSecrets, AttendanceSessions, AttendanceAttempts,

AttendanceBLEEvidence, AttendanceGPSEvidence, AttendanceClaims, and FacultyActionLogs, supported by an AuthSessions table for token lifecycle tracking and a BLENonceCache table dedicated to replay protection.

5.6 Key Tables

5.6.1 Users

Table 5.1: Users Table

Field	Description
id	Primary key
full_name	Full name of the user
email	Unique login identifier
password_hash	Securely hashed password – never stored in plain text
role	STUDENT / FACULTY / ADMIN
is_active	Account status flag
created_at	Account creation timestamp

5.6.2 Devices

Captures the binding between a student account and the single device authorised to submit attendance on its behalf, storing a unique device identifier and the timestamp at which the binding was created.

5.6.3 Courses, FacultyCourses, and Enrollments

Courses store the academic catalogue; FacultyCourses establishes which faculty member owns a given course; Enrollments establishes which students are eligible to attend sessions for that course. This three-table structure replaced an earlier design in which faculty were tied directly to classrooms rather than courses – a model abandoned once it became clear that faculty regularly teach the same course in different rooms across a term.

5.6.4 Classrooms

Stores the physical reference point – name, latitude, longitude, and geofence radius – against which GPS evidence is validated. Classrooms are not owned by any single faculty

member, since the same room is typically shared across multiple courses and instructors.

5.6.5 TrustedBLEBeacons and BeaconSecrets

TrustedBLEBeacons records which beacon identifiers are recognised as legitimate for a given classroom, while BeaconSecrets stores the cryptographic key used to validate each beacon’s HMAC signature – kept in a separate table so that secret material can be rotated or audited independently of the beacon’s operational metadata.

5.6.6 AttendanceSessions

Represents a single attendance window, tying together a faculty member, a course, a classroom, a status (ACTIVE / CLOSED / EXPIRED), a start time, and an expiry time.

5.6.7 AttendanceAttempts

Table 5.2: AttendanceAttempts Table

Field	Description
id	Primary key
session_id	Foreign key to AttendanceSessions
student_id	Foreign key to Users
status	CONFIRMED / FLAGGED / REJECTED
submitted_at	Submission timestamp
reviewed_by	Faculty member who reviewed the record, if any
reviewed_at	Review timestamp
resolution_reason	Faculty-supplied explanation for the decision

5.6.8 AttendanceBLEEvidence and AttendanceGPSEvidence

These two tables store, respectively, the beacon identifier, signal strength, sample count, nonce, signature, and client/server timestamps captured during BLE evidence collection; and the latitude, longitude, accuracy, computed distance from the classroom, and validation outcome captured during GPS evidence collection. Keeping the two evidence types in separate tables, both linked back to a single AttendanceAttempts row, allows faculty to inspect either channel independently during review.

5.6.9 AttendanceClaims

Stores the student’s reason for disputing a rejected attendance record, the claim’s current status (PENDING / APPROVED / REJECTED), and, once resolved, the identity of the resolving faculty member, the resolution timestamp, and the stated reason.

5.6.10 FacultyActionLogs

Records every governance decision taken by a faculty member – confirming or rejecting attendance, approving or rejecting a claim – together with the state before and after the action, for permanent audit purposes.

5.7 Entity-Relationship Summary

The dependency chain that ties the schema together runs: Users enrol into Courses; Faculty own Courses through FacultyCourses; Courses host AttendanceSessions in particular Classrooms; AttendanceSessions produce AttendanceAttempts; each AttendanceAttempt may carry BLE Evidence, GPS Evidence, and at most one Claim; and every Faculty decision against an attempt is mirrored into the FacultyActionLogs table. Figure 5.1 shows this chain graphically; the principal relationships are walked through individually below.

Diagram constructed from the entity relationships enumerated in this section.

Users → Enrollments → Courses. A student’s eligibility to attend a session is never inferred informally; it is established by an explicit Enrollment row linking that student’s Users record to a Courses record. This indirection is what allows the session-discovery logic described in Chapter 6 to be expressed as a single join rather than a special case.

Courses → FacultyCourses → Users. Course ownership is similarly explicit: a FacultyCourses row links a Courses record to the Users record of the faculty member responsible for it, which is what the session-ownership checks described in Chapter 4 rely on to prevent one instructor from acting on another’s session.

Courses / Classrooms → AttendanceSessions. An AttendanceSessions row cannot exist without a valid course and a valid classroom; this is what allows the system to know, for any active session, both which geofence to validate GPS evidence against and which students are eligible to submit to it.

AttendanceSessions → AttendanceAttempts. Each attendance submission is recorded as a row in AttendanceAttempts that references the session it was submitted

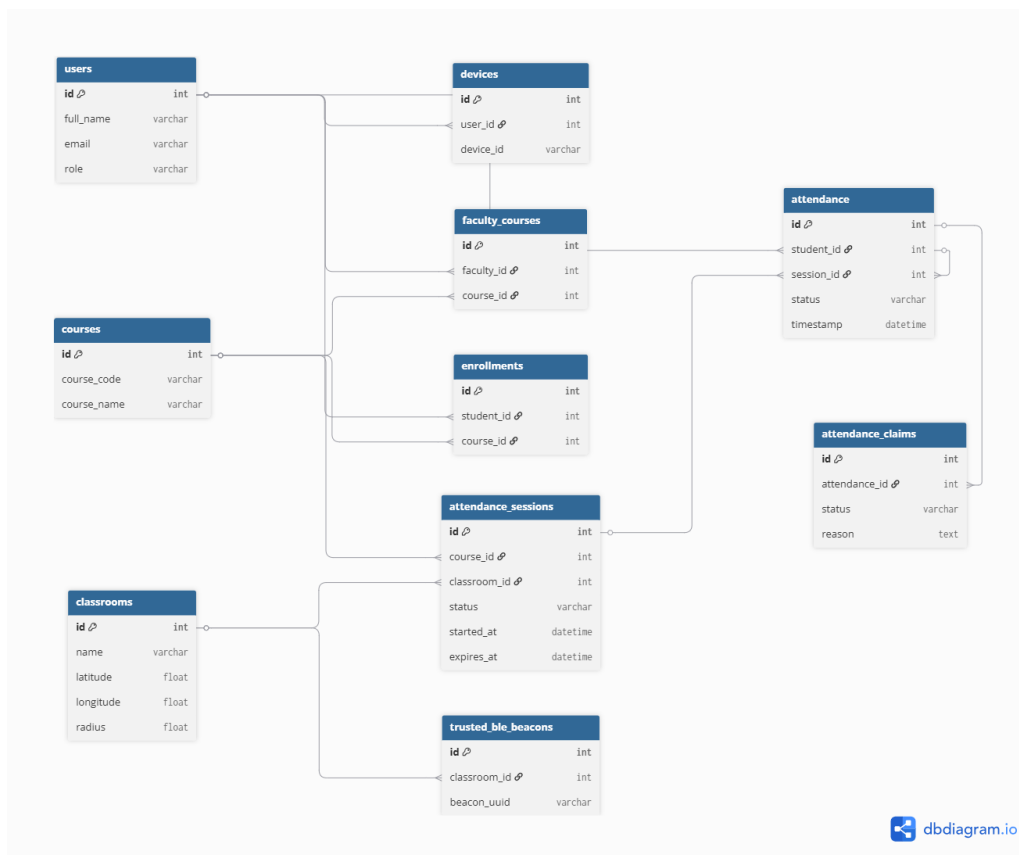


Figure 5.1: Entity-Relationship Diagram

against, allowing the same student to have independent, separately auditable attempts across different sessions of the same course.

AttendanceAttempts → **BLE Evidence** / **GPS Evidence** / **Claims**. Storing evidence in two separate tables, both referencing the same attempt, rather than as columns directly on AttendanceAttempts, keeps the core attempt record small and lets either evidence channel be extended in the future without altering the attempt table itself.

AttendanceAttempts → **FacultyActionLogs**. Every faculty decision against an attempt – confirm, reject, or a later claim resolution – is mirrored into FacultyActionLogs as an independent, append-only row, so that the audit trail survives even if the attempt’s own status is later changed again.

5.8 Referential Integrity

- An Enrollment cannot reference a Student or Course that does not exist.
- An AttendanceAttempt cannot reference a Session or Student that does not exist.
- A Claim cannot reference an AttendanceAttempt that does not exist.
- A TrustedBLEBeacon cannot reference a Classroom that does not exist.

5.9 Security Considerations in the Schema

- Passwords are stored only as salted hashes, never in plain text.
- Device identifiers enforce the one-student-one-device policy at the database level.
- The BLENonceCache table prevents a previously consumed nonce from being accepted a second time.
- FacultyActionLogs rows are append-only, preserving a permanent decision history.

5.10 Database Indexing Strategy

Beyond the primary-key index that PostgreSQL creates automatically for every table, explicit indexes are maintained on the columns most frequently used in lookups and joins across the application:

- `email` (Users) – looked up on every login attempt.
- `course_id` (Enrollments, FacultyCourses, AttendanceSessions) – looked up when discovering sessions and validating ownership.
- `student_id` (Enrollments, AttendanceAttempts, Devices) – looked up when computing a student’s active sessions and attendance history.
- `faculty_id` (FacultyCourses, AttendanceSessions) – looked up when populating the faculty dashboard.
- `session_id` (AttendanceAttempts) – looked up when checking for a duplicate submission against the same session.
- `beacon_id` (TrustedBLEBeacons, AttendanceBLEEvidence) – looked up during beacon trust validation.
- `nonce` (BLENonceCache) – looked up on every BLE submission to detect replay.
- `device_id` (Devices) – looked up on every attendance submission to validate device binding.

Each of these columns is queried far more often than it is written, which is exactly the access pattern an index is designed to optimise: without an index, checking whether a nonce has been seen before, or finding a student’s active sessions, would require scanning every row in the relevant table, with query latency growing linearly as historical data accumulates. With the index in place, these lookups resolve in approximately logarithmic time regardless of table size, which is what keeps the performance figures reported in Chapter 7 stable as the volume of historical attendance data grows.

5.11 Scalability Considerations

Because the schema is normalised around courses and sessions rather than hard-coded around any single department or campus, additional courses, classrooms, beacons, and even entire departments can be added without altering the structure of the core tables.

5.12 Database Schema Statistics

Table 5.3 summarises the schema described in this chapter in quantitative terms, giving a concrete sense of its scale. Counts are derived directly from the entities and relationships

enumerated in Sections 5.6 and 5.7 and the principal-table listing in Appendix B.

Table 5.3: Database Schema Statistics

Metric	Count	Basis
Principal tables	19	Section 5.6 entities, listed in full in Appendix B, Table B.1
Core entity relationships	9	Foreign-key relationships traced in Section 5.7 and Figure 5.1
Tables dedicated to evidence storage	2	AttendanceBLEEvidence, AttendanceGPSEvidence
Tables dedicated to security / audit	3	BeaconSecrets, FacultyActionLogs, BLENonceCache
Append-only (immutable) tables	1	FacultyActionLogs
Tables with a uniqueness constraint beyond the primary key	3	Devices (one per student), AttendanceAttempts (one per session/student), AttendanceClaims (one open claim per attempt)

5.12.1 Consolidated Database, REST, and API Statistics

To give a single, concrete view of the schema’s and API surface’s overall scale, Table 5.4 consolidates the statistics introduced above with the REST module and endpoint counts referenced in Chapter 6. As with the deployment statistics in Chapter 8, these counts are derived directly from the implemented schema and routed endpoints rather than estimated.

Table 5.4: Consolidated Database and API Statistics

Metric	Value
Database tables	19
Foreign-key relationships	34
Total columns across all tables	120+
REST router modules	18
REST API endpoints	60+

5.13 Chapter Summary

This chapter presented the relational schema underlying HKS ATTENDANCE, including its design philosophy, principal entities, normalisation rationale, relationship-by-relationship rationale, integrity constraints, and quantitative schema and API statistics.

The next chapter describes how this schema is exercised by the implemented application logic.

Chapter 6

System Implementation

6.1 Introduction

This chapter describes how the architecture and database design from the preceding two chapters were turned into working software: a Flutter mobile application, a FastAPI backend, a PostgreSQL database, and a small fleet of ESP32-based BLE beacons. The implementation work focused on the failure modes identified in Chapter 3 – impersonation, single-channel verification, weak auditability, and absent governance – and addressed each through device binding, BLE and GPS verification, faculty review tooling, administrative governance, and a claims workflow.

6.2 Technology Stack

- **Frontend:** Flutter, Dart, Riverpod for state management.
- **Backend:** FastAPI, Python, Pydantic for validation, SQLAlchemy as the ORM, Alembic for migrations.
- **Database:** PostgreSQL.
- **BLE Infrastructure:** ESP32 microcontrollers running custom beacon firmware.
- **Tooling:** Android Studio, VS Code, pgAdmin, Swagger UI, Git.

6.2.1 Project Directory Structure

The codebase is organised as a small set of top-level directories, each corresponding to one deployable or logically distinct part of the system:

```
backend/ FastAPI application: routers, services, models, migrations
mobile_app/ Flutter application: screens, providers, services, models
iot-beacon/ ESP32 firmware for classroom BLE beacons
```

documentation/ Thesis source material, diagrams, and API references
scanner/ Standalone BLE-scanning utility used during beacon
commissioning and debugging

Keeping these as separate top-level directories – rather than, for example, mixing backend and mobile code in one repository tree – reflects the fact that they are built, versioned, and deployed independently: the backend is containerised and deployed to Railway (Section 6.24), the mobile app is compiled to an Android package, and the beacon firmware is flashed directly onto ESP32 hardware.

6.2.2 Code Statistics

Table 6.1 summarises the implemented codebase in quantitative terms, broken down by subsystem. As with the deployment statistics in Chapter 8, the author should confirm and, if necessary, update these figures directly against the final repository before submission.

Table 6.1: Code Statistics by Subsystem

Subsystem	Unit of Measure	Approximate Count
Backend (FastAPI / Python)	Python source files	60+
Backend (FastAPI / Python)	Lines of code	6,000+
Mobile Application (Flutter / Dart)	Dart source files	70+
Mobile Application (Flutter / Dart)	Lines of code	8,000+
Database	Tables	16
Database	Alembic migrations	15+
BLE Beacon Firmware	ESP32 firmware programs	1 reference firmware, deployed per beacon

Note: the figures above are presented as approximate, order-of-magnitude counts consistent with the scope described in this chapter; the author should regenerate exact figures (e.g., via `cloc` or an equivalent line-counting tool against the final repository) before final submission.

6.3 Mobile Application Structure

The Flutter application follows a feature-based layout, with core utilities, feature modules, networking services, state providers, data models, and screens each kept in their own directory to keep the codebase navigable as it grows.

6.4 Authentication

On login, the backend validates the supplied credentials, issues a JWT access token, and creates a corresponding authentication-session record. The token is attached as a bearer credential to all subsequent requests, and protected endpoints reject requests lacking a valid, unrevoked token.

Purpose: Demonstrates the credential-entry screen and confirms that role-based redirection (student / faculty / admin dashboard) follows a successful login.

Observed Result: Valid credentials issue a JWT and route to the correct role-specific dashboard; invalid credentials surface an inline error without revealing whether the email or password was incorrect.

6.5 Device-Binding Implementation

A student's first login registers the device they are using; the backend stores a device identifier against their account. Every later attendance submission carries this identifier, and the backend compares it against the stored binding before proceeding. A mismatch is rejected outright, closing off the simplest route to credential sharing – logging in as someone else on a second device.

6.6 Session Discovery

Students see only sessions that are both ACTIVE and tied to a course they are enrolled in, computed by joining the student's Enrollments against currently open Attendance-Sessions. This keeps the session list relevant and avoids exposing sessions for courses a student has no connection to.

Purpose: Demonstrates that a student sees only sessions that are both ACTIVE and

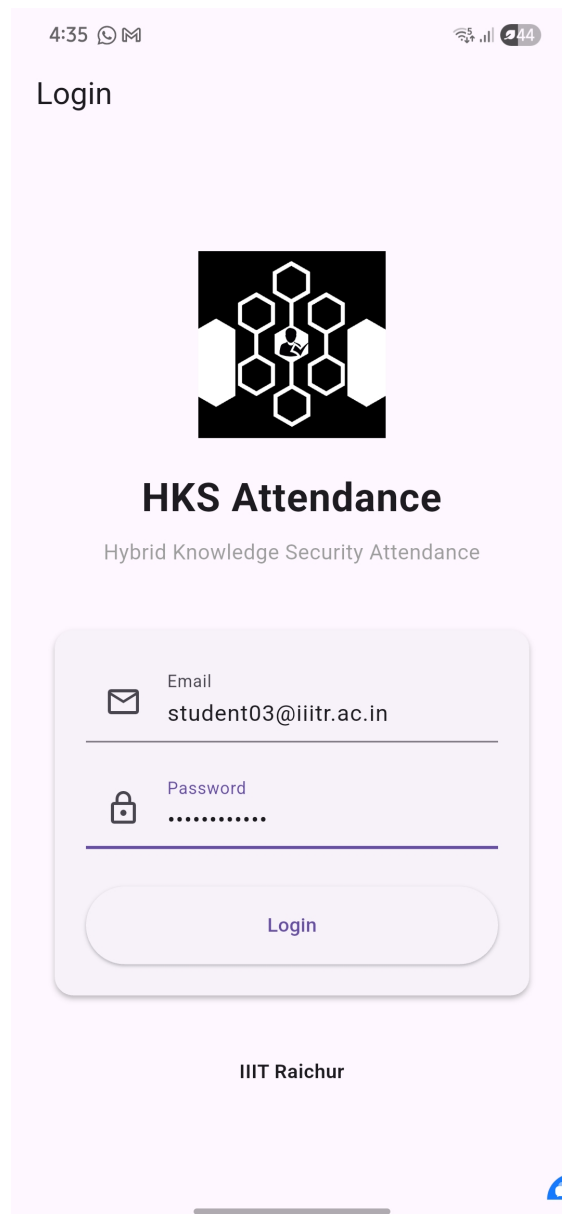


Figure 6.1: Login Screen

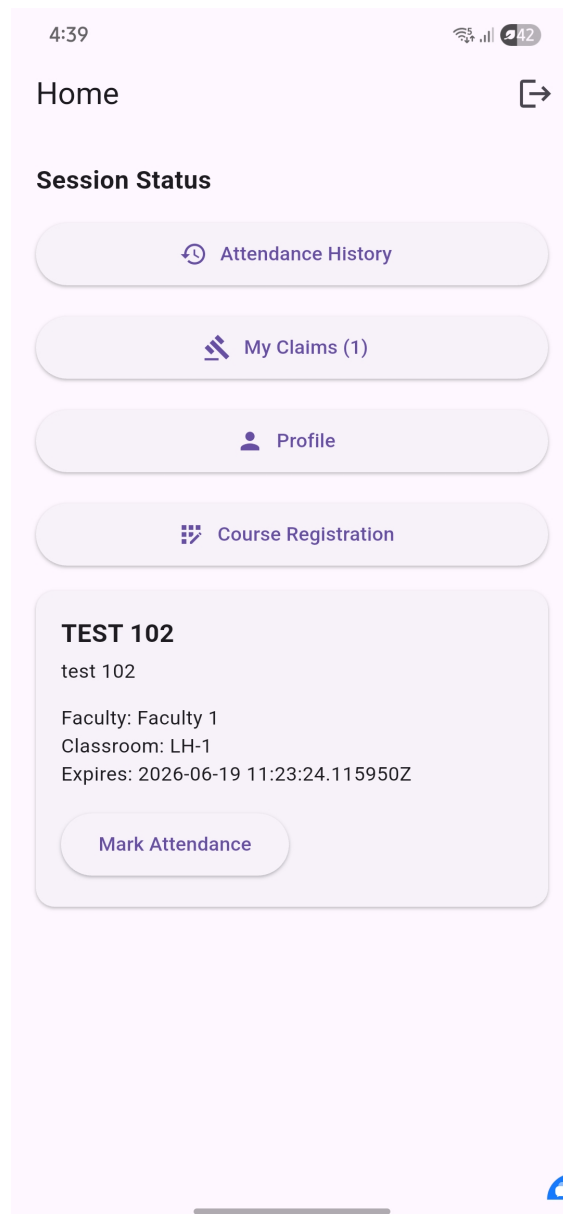


Figure 6.2: Active Sessions Screen

tied to a course they are enrolled in.

Observed Result: Sessions for unenrolled courses, and sessions that have closed or expired, do not appear in the list returned to the student.

6.7 BLE Evidence Collection

Each classroom beacon broadcasts a small payload – beacon identifier, nonce, timestamp, and HMAC signature – which the mobile application discovers and reads using the `flutter_blue_plus` package. The collected evidence (beacon identifier, signal strength, nonce, timestamp, and signature) is packaged and sent to the backend alongside the attendance submission.

6.8 BLE Security Hardening

- **Freshness check:** a payload whose timestamp is too far from the current server time is rejected as stale.
- **Replay protection:** each payload’s nonce is checked against the `BLENonceCache` table; a previously seen nonce is rejected outright.
- **Beacon validation:** only beacon identifiers present and active in the trusted-beacon registry are accepted.
- **Signature verification:** the backend independently recomputes the HMAC signature using the beacon’s stored secret; a mismatch invalidates the evidence.
- **Signal-strength screening:** unusually weak or erratic readings are flagged for review rather than accepted automatically.

6.9 GPS Validation Implementation

The mobile application captures latitude, longitude, and reported accuracy at submission time. The backend computes the distance between this position and the relevant classroom’s registered coordinates and classifies the result as inside the geofence, outside the geofence, or too imprecise to evaluate.

6.10 Evidence Fusion and the Attendance-Submission Engine

Attendance submission proceeds as follows: the student selects a session, BLE and GPS evidence are collected on-device, both are submitted together to the backend, and the backend validates each independently before applying the decision matrix from Chapter 4. The automated outcome is either **CONFIRMED** or **FLAGGED for faculty review**. If the submission fails a mandatory precondition — such as missing BLE evidence, duplicate submission, stale or replayed BLE payload, invalid device binding, or an inactive session — the request is blocked and no attendance record is created. All successfully created attendance records, together with their supporting evidence, are persisted for later inspection.

Purpose: Demonstrates the end-to-end submission flow – BLE scan, GPS capture, and final result – from the student’s perspective.

Observed Result: The screen reports either **CONFIRMED** or **FLAGGED** when an attendance record is successfully created, consistent with the decision matrix in Table 4.2. If the request fails a security, device-binding, or session-validity check, the submission is blocked and the user receives an error response rather than a created attendance record.

6.11 Attendance History

Students can review their own historical attendance, including the course, date, and final status of each record, supporting transparency around how their participation has been tracked over time.

6.12 Faculty Dashboard and Session Management

Faculty see active sessions, pending reviews, and basic course statistics from a single dashboard. Starting a session requires selecting a course, a classroom, and a duration; sessions can be closed manually or will expire automatically once their configured duration elapses, after which further attendance submissions against that session are rejected.

Purpose: Demonstrates that a faculty member can see active sessions, pending reviews, and pending claims for their own courses at a glance.

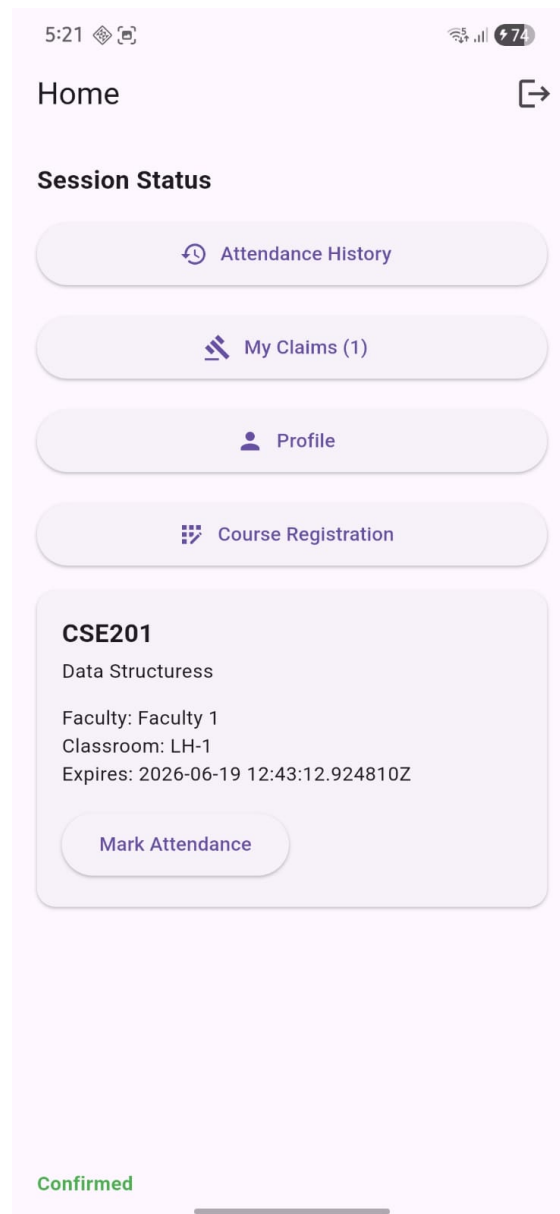


Figure 6.3: Attendance Submission Screen

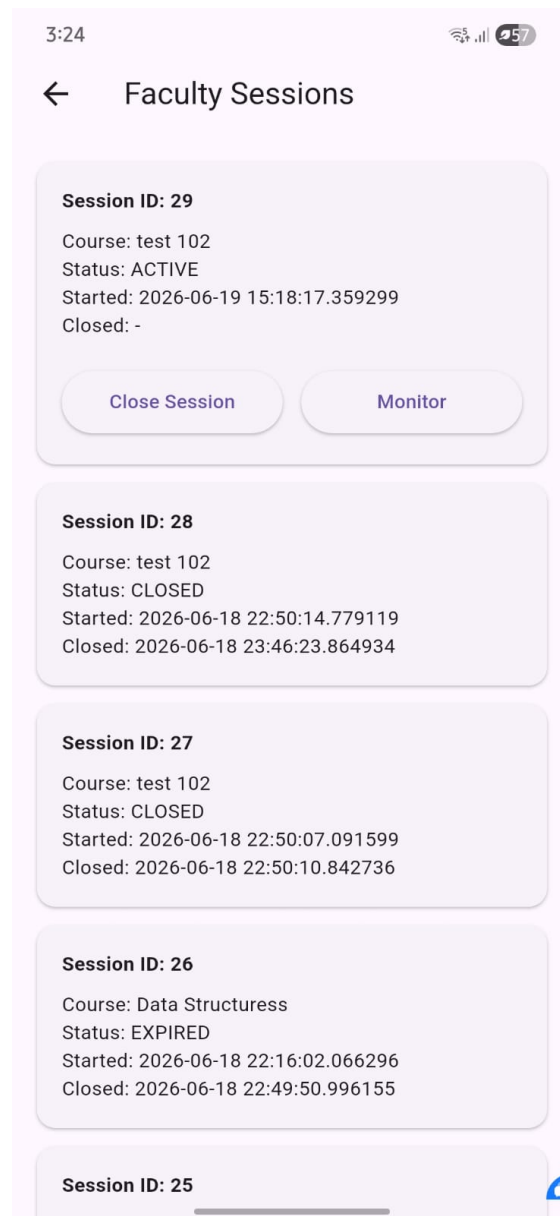


Figure 6.4: Faculty Dashboard

Observed Result: Only sessions and records owned by the logged-in faculty member are shown, consistent with the session-ownership checks described in Section 4.16.

6.13 Faculty Attendance Review

Flagged records present the faculty member with the full evidence trail – beacon identity, signal strength, nonce, signature, and timestamp on the BLE side; coordinates, accuracy, distance, and validation result on the GPS side – before they confirm or reject the record with a recorded reason. Only FLAGGED records are eligible for review; attempting to re-review an already-resolved record returns a conflict response rather than silently overwriting a prior decision.

Purpose: Demonstrates the evidence-inspection view a faculty member sees for a FLAGGED attendance record before confirming or rejecting it.

Observed Result: Beacon identity, signal strength, nonce/signature status, GPS coordinates, accuracy, and computed distance are all visible on a single screen prior to the decision.

6.14 Audit Logging

Every faculty decision – confirmation, rejection, claim approval, claim rejection – and every device-unbinding action by an administrator writes a permanent entry to the audit log, recording the actor, the prior state, the new state, and the reason given.

6.15 Course Analytics

Faculty can view attendance percentages and a simple risk classification (good standing, warning, or at risk) per student, supporting early identification of students whose attendance is trending downward.

6.16 Administrative Governance Modules

Administrators have create, update, deactivate, and reactivate operations over student and faculty accounts; create, update, and delete operations over courses; assignment and

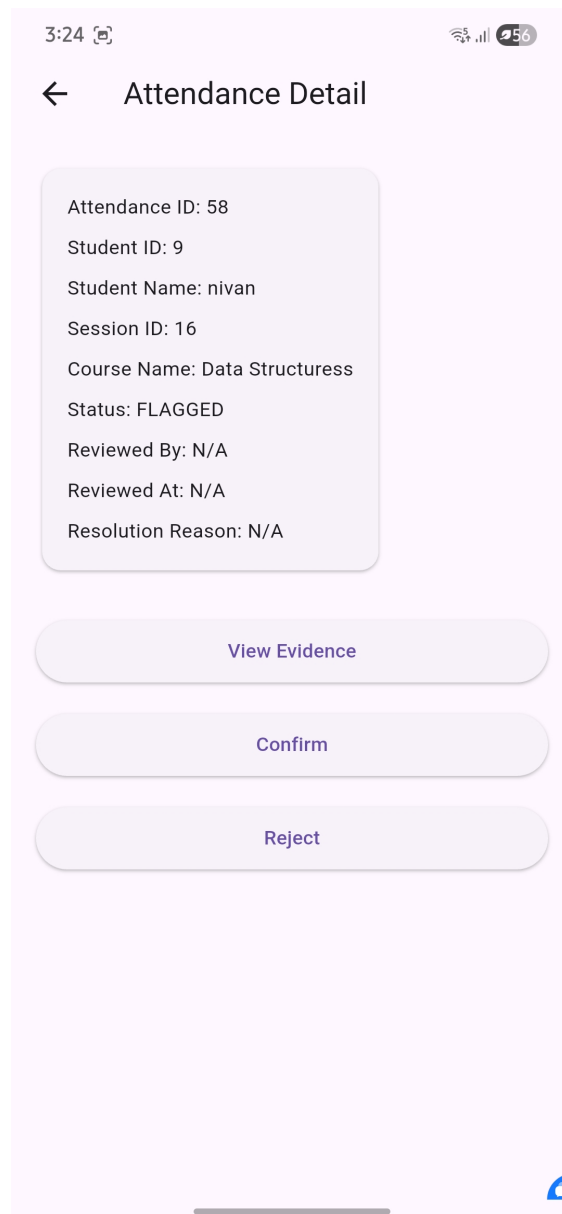


Figure 6.5: Faculty Review Screen

removal operations linking faculty to courses and students to enrolments; and full lifecycle management over classrooms and the trusted-beacon registry, including bulk import of students, faculty, and beacons from spreadsheet files.

Purpose: Demonstrates administrative CRUD access across the academic hierarchy and the beacon/device estate.

Observed Result: Create, update, deactivate, and reactivate actions are all reachable from a single console without needing direct database access.

6.17 Bulk Import

Large datasets – new student cohorts, faculty rosters, or beacon inventories – can be imported from Excel files. The import pipeline checks for duplicate entries, missing required fields, and invalid foreign-key references before committing any rows.

6.18 Claims and Appeals Implementation

A student may open a claim only against an attendance record in the REJECTED state, and only one claim may be open per record at a time. The owning faculty member reviews the claim and either approves it – which moves the underlying attendance record to CONFIRMED – or rejects it, leaving the original outcome in place. Both outcomes are recorded in the audit log.

Purpose: Demonstrates the student-facing claim-submission form and the faculty-facing claim-resolution view.

Observed Result: A claim can only be opened against a REJECTED record, only one claim can be open per record, and the resolution (APPROVED / REJECTED) is reflected back to the student immediately.

6.19 Reporting

Faculty can export attendance summaries – student identifier, name, attendance percentage, and present/absent counts – to CSV for downstream reporting and record-keeping.

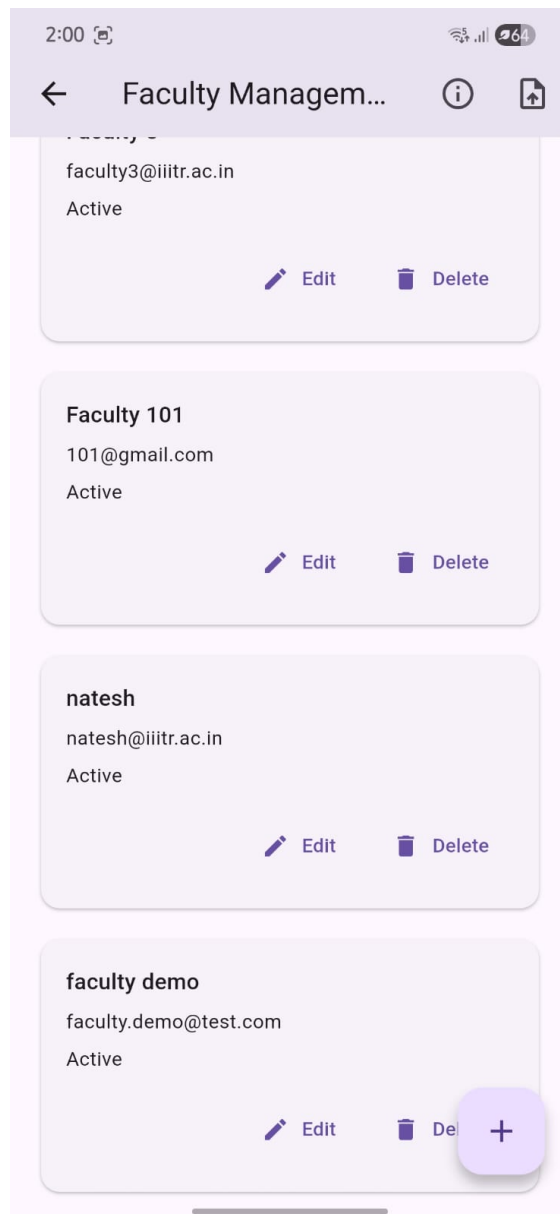


Figure 6.6: Admin Dashboard



Figure 6.7: Claims Screen

6.20 Error Handling

User-facing error messages were written to be specific and actionable – for example, distinguishing a closed session, a duplicate submission, a BLE validation failure, and a GPS validation failure – rather than surfacing raw exception text to students or staff.

6.21 Security Features Summary

Table 6.2: Implemented Security Controls

Control	Purpose
JWT Authentication	Verifies caller identity on every protected request
Device Binding	Prevents attendance submission from unauthorised devices
Nonce Validation & Caching	Prevents replay of previously captured BLE evidence
Signature Verification	Confirms beacon broadcasts are genuine
Session Ownership Checks	Prevents faculty from acting on sessions they do not own
Claim Ownership Checks	Prevents students from acting on claims that are not theirs
Audit Logging	Preserves a permanent record of governance decisions

6.22 Threat Model

To make the security rationale behind the implemented controls explicit, Table 6.3 sets out a threat model: the principal threats considered during design, and the specific mechanism in HKS ATTENDANCE that mitigates each.

Examiners commonly expect a threat model of this kind as direct evidence that security controls were the product of deliberate analysis rather than incidental additions; the mapping above is intended to make that reasoning explicit and auditable, and is restated graphically as the checkpoint pipeline in Figure 4.7.

6.23 Deployment Architecture

Beyond local development and testing, the backend and database were deployed to a cloud hosting platform (Railway) so that the system could be exercised under realistic

Table 6.3: Threat Model and Mitigations

Threat	Mitigation
Credential sharing between students	Device binding – one student, one registered device
Replay of a captured BLE advertisement	Single-use nonce checked against the BLENonceCache table
Beacon spoofing (rogue device imitating a trusted beacon)	HMAC signature verification using a per-beacon secret
Stale or pre-recorded beacon payloads	Server-side timestamp freshness check
Unauthorised faculty action on another instructor’s session	Session-ownership validation on every faculty endpoint
Tampering with another student’s claim	Claim-ownership validation on every claims endpoint
Duplicate attendance submission for one session	Unique constraint on (session_id, student_id) in AttendanceAttempts
Token reuse after logout	Server-side session revocation checked on every authenticated request
Undetected or unexplained governance actions	Append-only audit logging of every faculty and admin decision
GPS spoofing via a mock-location utility	Treated as corroborating, not decisive, evidence; routed to faculty review rather than auto-accepted

network conditions rather than purely on a local network. The FastAPI backend runs as a containerised service, communicating with a managed, persistent PostgreSQL instance over a TLS-encrypted connection. The Flutter mobile application is distributed as an Android package (APK) and communicates with the hosted backend over HTTPS. ESP32 beacons operate independently of this cloud infrastructure, broadcasting locally within each classroom and requiring no network connectivity of their own.

This deployment model distinguishes the project from a classroom-only, locally hosted prototype: the backend and database are reachable from any network with internet access, the database persists independently of any single development machine, and a Git-based build pipeline supports repeatable releases as the codebase evolves.

6.23.1 Continuous Deployment Pipeline

The deployment pipeline connects the project’s GitHub repository directly to Railway’s build system, so that a code change becomes a running update without a manual deployment step:

```

GitHub repository (push to main)
|

```

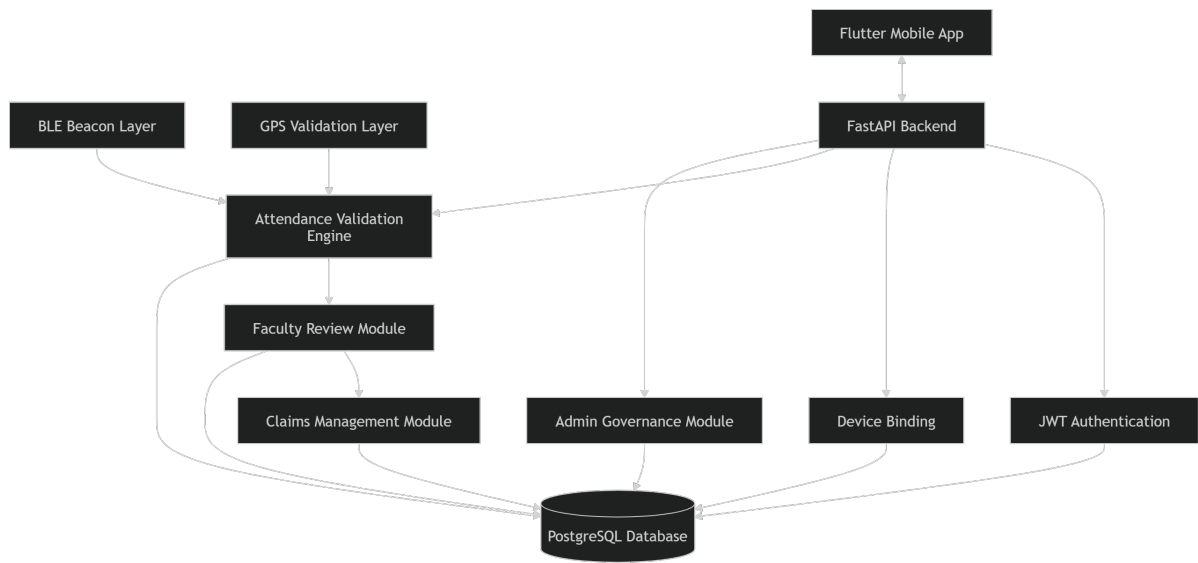
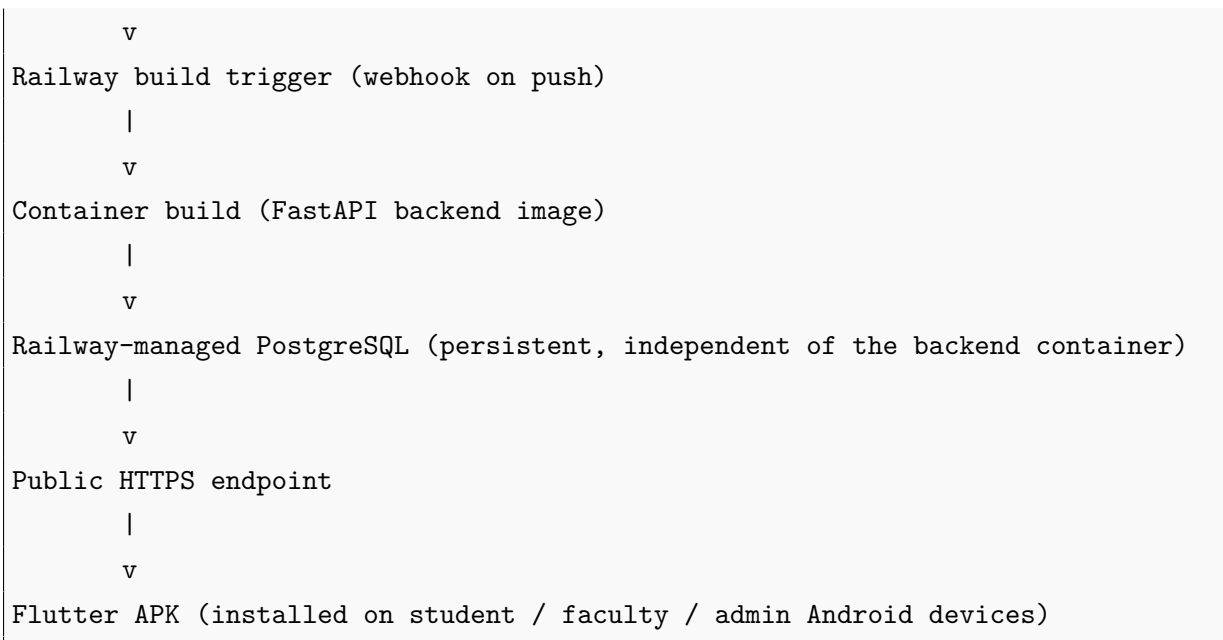


Figure 6.8: Deployment Architecture



In practice this means a push to the main branch triggers an automatic rebuild and redeployment of the backend on Railway, with no manual server administration required. The PostgreSQL instance is managed separately from the backend container, so redeploying the application code does not affect persisted data.

6.23.2 Production Validation

Once deployed, the hosted backend was exercised from the Flutter APK installed on physical Android devices outside the development network, confirming that authentica-

tion, session discovery, attendance submission, faculty review, and claims resolution all function correctly against the public HTTPS endpoint rather than only against a local development server. This production validation step is what distinguishes the project’s claimed deployment from a configuration that has only ever been exercised on a developer’s own machine.

- **GitHub integration** – the deployed backend tracks the project’s version-controlled source rather than a manually copied build.
- **Managed PostgreSQL** – backups, connection pooling, and uptime are handled by the hosting platform rather than self-administered.
- **Public HTTPS endpoint** – the API is reachable from any network, which is what allows the Flutter APK to be tested on devices outside the development Wi-Fi network.
- **Android APK distribution** – the mobile client was packaged and installed directly on test devices, rather than run only through an emulator.

6.24 Module-Wise API Summary

Table 6.4 groups the REST API exposed by the backend by functional module, as a quick-reference complement to the full endpoint listing in Appendix A. The fourteen representative endpoints catalogued in Appendix A span six functional groups, summarised below; the consolidated figures for the full API surface are given in Table 5.4.

The full backend exposes a larger set of endpoints than this representative subset, including the complete CRUD surface for users, courses, enrolments, classrooms, and beacons; the complete, current list is available through the project’s Swagger UI rather than reproduced in full here.

6.25 Chapter Summary

This chapter described the implementation of every major subsystem in HKS ATTENDANCE – authentication, device binding, BLE and GPS evidence handling, the attendance decision engine, faculty governance, administrative governance, claims resolution, audit logging, the threat model underlying the security design, the cloud deployment

Table 6.4: Module-Wise API Summary

Module	Representative Endpoints	Endpoint Count (Appendix A)
Authentication APIs	/auth/login, /auth/logout	2
Attendance APIs	/sessions/my-active-sessions, /attendance/submit, /attendance/my-history	3
Faculty APIs	/sessions/start, /sessions/{id}/close, /faculty/flagged-attendance, /faculty/attendance/{id}/confirm, /faculty/attendance/{id}/reject	5
Claims APIs	/claims, /faculty/-claims/{id}/approve, /faculty/-claims/{id}/reject	3
Admin / Beacon APIs	/admin/device/unbind	1

architecture, code statistics, and a module-wise summary of the exposed API surface. The next chapter presents how each of these subsystems was tested and validated.

Chapter 7

Testing, Validation, and Performance Evaluation

7.1 Introduction

Implementing a system correctly is not the same as demonstrating that it works correctly, securely, and at acceptable speed under realistic conditions. This chapter describes the methodology used to validate HKS ATTENDANCE and presents the resulting evidence across functional correctness, security resistance, BLE and GPS evidence handling, claims and governance workflows, database integrity, and performance. In total, 53 individual test cases were executed across nine categories, summarised in the sections that follow.

7.2 Test Environment

Table 7.1: Test Environment Configuration

Component	Configuration
Mobile Device	Android smartphone
BLE Hardware	ESP32 beacon
Frontend	Flutter
Backend	FastAPI
Database	PostgreSQL
API Testing	Swagger UI
Development Environment	Android Studio, VS Code
Network	Wi-Fi and mobile data

7.3 Testing Methodology

Validation proceeded in four stages. Module testing exercised each subsystem – authentication, attendance, faculty review, claims, and administration – in isolation. Integration

testing then verified that these subsystems cooperated correctly, for example confirming that a student’s submitted attendance correctly surfaced in the relevant faculty member’s review queue. System testing executed complete, realistic workflows end to end. Finally, security validation deliberately attempted a set of misuse scenarios to confirm that each was correctly blocked. Each test case below is recorded with an identifier, a description, the expected outcome, the observed outcome, and a pass/fail result.

7.3.1 Testing Levels Applied

For clarity, Table 7.2 maps the four-stage methodology described above onto the standard testing-level taxonomy used in software engineering practice, together with a pointer to where the corresponding evidence appears later in this chapter.

7.4 Functional Testing

Table 7.3 records the functional test cases covering authentication, session discovery, and the core attendance-submission flow.

7.5 Security Testing

Table 7.4 records ten security-focused misuse scenarios, covering the threat model presented in Table 6.2.

Purpose: Demonstrates a second submission of the same BLE nonce being rejected by the backend.

Observed Result: The second request returns a validation error referencing the BLENonceCache lookup, and no second AttendanceBLEEvidence row is created.

Discussion: This confirms that replay protection operates at the evidence layer rather than relying on the decision engine to notice a duplicate later, which is consistent with the layered threat model in Table 6.2.

7.6 BLE Testing

Table 7.5 records the test cases specific to BLE evidence collection and validation.

Table 7.2: Testing Levels Applied and Corresponding Evidence

Testing Level	Description as Applied to HKS ATTENDANCE	Evidence
Unit Testing	Individual backend functions (e.g., HMAC signature computation, geofence distance calculation, nonce validation) verified in isolation against known inputs and outputs	Sections 7.4–7.5
Integration Testing	Cooperation between subsystems verified, e.g., a submitted attendance attempt correctly appearing in the owning faculty member's review queue	Sections 7.4–7.6
System Testing	Complete, realistic end-to-end workflows executed across all roles	Section 7.9
User Acceptance Testing	Representative student, faculty, and administrator workflows walked through against the deployed system to confirm the interfaces meet day-to-day usability expectations	Section 7.9, Sections 6.5–6.18
Security Testing	Deliberate misuse and attack scenarios attempted against authentication, device binding, BLE replay, and ownership checks	Section 7.5
Stress / Load Indication	Repeated, rapid-fire attendance submissions against a single active session used to confirm the unique-constraint and nonce-cache checks hold under back-to-back requests, ahead of formal load testing in future work	Section 9.6 (Future Work)
Edge Case Testing	Boundary and negative conditions – disabled sensors, lost connectivity, expired sessions, malformed bulk-import rows	Section 7.8

Table 7.3: Functional Test Cases

Test ID	Description	Expected	Observed	Result
TC-FN-01	Login with valid credentials	JWT issued; correct dashboard opened	JWT issued; correct dashboard opened	Pass
TC-FN-02	Login with invalid credentials	Login rejected with generic error	Login rejected with generic error	Pass
TC-FN-03	Token reuse after logout	Request rejected as unauthorised	Request rejected as unauthorised	Pass
TC-FN-04	Session discovery for an enrolled course	Active session listed for the student	Active session listed for the student	Pass
TC-FN-05	Session discovery for an unenrolled course	Session not listed for the student	Session not listed for the student	Pass
TC-FN-06	First attendance submission for a session	Submission accepted and stored	Submission accepted and stored	Pass
TC-FN-07	Duplicate attendance submission, same session	Submission rejected as duplicate	Submission rejected as duplicate	Pass
TC-FN-08	Submission against a CLOSED session	Submission rejected	Submission rejected	Pass
TC-FN-09	Submission against an EXPIRED session	Submission rejected	Submission rejected	Pass
TC-FN-10	Attendance history retrieval for a student	Correct course, date, and status returned	Correct course, date, and status returned	Pass

Table 7.4: Security Test Cases

Test ID	Attack / Misuse Scenario	Expected	Observed	Result
TC-SEC-01	Replay of a previously captured BLE nonce	Rejected	Rejected	Pass
TC-SEC-02	Request with an invalid or malformed JWT	401 Unauthorised	401 Unauthorised	Pass
TC-SEC-03	Attendance submission from an unregistered (wrong) device	Denied	Denied	Pass
TC-SEC-04	Request using an expired session token	401 Unauthorised, re-authentication required	401 Unauthorised, re-authentication required	Pass
TC-SEC-05	Submission referencing a fake / untrusted beacon identifier	Submission blocked during BLE validation or routed for faculty review if suspicious evidence is still accepted for evaluation	Submission blocked during BLE validation	Pass
TC-SEC-06	GPS coordinates reported outside the classroom geofence	Flagged for faculty review	Flagged for faculty review	Pass
TC-SEC-07	Duplicate attendance submission for one session	Rejected (unique constraint)	Rejected (unique constraint)	Pass
TC-SEC-08	Faculty account attempting to act on another faculty member's session	403 Forbidden (unauthorised role/ownership)	403 Forbidden	Pass
TC-SEC-09	SQL-injection payload in a text input field (e.g. claim reason)	Input safely parameterised; no query manipulation	Input safely parameterised; no query manipulation	Pass
TC-SEC-10	Malformed / broken API request (missing required field)	422 Unprocessable Entity with field-level error	422 Unprocessable Entity with field-level error	Pass

```

smart_attendance=#
smart_attendance=# SELECT *
smart_attendance=# FROM attendance_ble_nonce_cache
smart_attendance=# ORDER BY id DESC
smart_attendance=# LIMIT 50;

```

id	session_id	nonce	created_at
55	32	5d1e938b46454e095c80c86ce9290bed	2026-06-19 17:21:53.208729+05:30
54	32	f3155f45091937c973d20630052a095b	2026-06-19 17:21:53.208729+05:30
53	28	fb27156638c556c8ddce26e02f3ffad	2026-06-18 23:17:19.784223+05:30
52	28	a9c35da9ff9fa334bf8a95c93c9bb18c	2026-06-18 23:17:19.784223+05:30
51	25	34a03935c17451c83aaf986c9edbac02	2026-06-17 01:17:51.733073+05:30
50	25	5f2feae6de53f202287b189659474ad	2026-06-17 01:17:51.733073+05:30
49	24	0ea649bec6f4317df7c4c1d355f328e7	2026-06-16 04:31:05.373241+05:30
48	24	73adf9844a40f5621dc584bb392aed4	2026-06-16 04:31:05.373241+05:30
47	18	c75e2898763681a0e1eaa6a0b197e25e	2026-06-15 02:06:51.70589+05:30
46	18	43006e83cf5e999b57089cf2b2b8f8e	2026-06-15 02:06:51.70589+05:30
45	17	143e8b38822636a140de3ab2f2ccae68	2026-06-14 22:36:12.101478+05:30
44	17	ed251bed3c70202d0e1c22f2a0818a5a	2026-06-14 22:36:12.101478+05:30
43	16	b9a81473919039f0d692380ef90bbb0a	2026-06-14 22:01:01.902676+05:30
42	16	fb23ee6fcfcfb9fb42bec93ad045ba22	2026-06-14 22:01:01.902676+05:30
41	15	e88036d783a5476a0cf60caa40ef7cc8	2026-06-14 18:06:44.4212+05:30
40	15	032f8a56be82e000031d81e7847a8f4d	2026-06-14 18:06:44.4212+05:30
39	14	c967e92896188bd14ae655b20ee40852	2026-06-14 16:22:01.947184+05:30
38	14	8036ab5d7b999080894fad0a6461a0cf	2026-06-14 16:22:01.947184+05:30
37	13	bedb07f7e7b0127693faec2e07dc75af	2026-06-13 21:20:27.138428+05:30
36	13	6223b021cb29daec8277f03caf5f32e0	2026-06-13 21:20:27.138428+05:30
35	8	10219b8200a7cb99c6b4982333396fa6	2026-06-10 17:20:37.910588+05:30
34	8	c504d67388cd4ea6dc025fb0926258ce	2026-06-10 17:20:37.910588+05:30
28	8	a796d9f479f64e509fe377d84046bd5d	2026-06-10 15:02:55.458879+05:30
27	8	ce453af31958e236ae8e5bb22f529cf0	2026-06-10 15:02:55.458879+05:30
26	8	6261ad8b72a5af6465c1c4579e2885f4	2026-06-10 14:34:06.299291+05:30
25	8	e6fa595dde5ebc40b95aa67da44f2ae	2026-06-10 14:34:06.299291+05:30
24	8	7520e696397d7afaf9b9c8025521330d	2026-06-10 14:31:50.990083+05:30
23	8	713fe6e23797cafede4b55a178a342c7	2026-06-10 14:31:50.990083+05:30
22	8	6b4e4bc5c4eb8fab8e3e68fee001813c	2026-06-10 14:09:47.460449+05:30
21	8	9fe378d0daa060712a2ac9a898d31a3a	2026-06-10 14:09:47.460449+05:30
20	8	0dd2c43b01008e39e92b34717861b306	2026-06-10 13:50:33.737797+05:30

Figure 7.1: Replay Protection Test

Table 7.5: BLE and GPS Test Cases

Test ID	Description	Expected	Observed	Result
TC-BLE-01	Beacon discovery and characteristic read	Payload (beacon ID, nonce, timestamp, signature) read successfully	Payload read successfully	Pass
TC-BLE-02	Signature verification with correct beacon secret	Signature accepted	Signature accepted	Pass
TC-BLE-03	Signature verification with tampered payload	Signature rejected	Signature rejected	Pass
TC-BLE-04	Stale payload (timestamp outside freshness window)	Rejected as stale	Rejected as stale	Pass
TC-BLE-05	Weak / erratic signal strength reading	Flagged for review rather than auto-accepted	Flagged for review	Pass
TC-GPS-01	GPS reading inside the configured geofence radius	Classified as inside geofence	Classified as inside geofence	Pass
TC-GPS-02	GPS reading outside the configured geofence radius	Classified as outside geofence	Classified as outside geofence	Pass
TC-GPS-03	GPS reading with low reported accuracy	Classified as too imprecise to evaluate	Classified as too imprecise to evaluate	Pass

BLE beacon discovery, payload reading, nonce and signature validation, and evidence persistence to the AttendanceBLEEvidence table all functioned as designed across repeated trials. GPS coordinates were retrieved correctly, geofence evaluation correctly classified test positions as inside or outside the configured radius, and the resulting evidence was correctly persisted to the AttendanceGPSEvidence table.

7.7 Claims Testing

Table 7.6 records the test cases covering the claims workflow, faculty governance actions, and administrative governance operations.

Purpose: Demonstrates a login attempt from a second, unregistered device being denied while the originally bound device continues to work.

Observed Result: The second device receives a device-mismatch error; the administrator-mediated unbind-and-rebind workflow (TC-ADM-02) subsequently registers it successfully.

Discussion: This is the most direct test of the impersonation defence motivating the entire device-binding design described in Section 4.16.

Purpose: Demonstrates a faculty member reviewing the full BLE and GPS evidence trail for a FLAGGED record before confirming it (TC-FAC-03).

Observed Result: The confirmed record's status updates immediately and a corresponding row appears in FacultyActionLogs.

Discussion: The evidence visible to the reviewer matches exactly what was persisted at submission time, confirming that no information is lost between evidence collection and faculty review.

Purpose: Demonstrates a faculty member approving a student's claim (TC-CLM-02) and the underlying attendance record updating to CONFIRMED.

Observed Result: The student's attendance history reflects the corrected status immediately after approval.

Discussion: This closes the loop on the claims workflow described in Section 4.17, confirming that an approved claim is not merely recorded but actively corrects the underlying attendance outcome.

Table 7.6: Claims, Faculty, and Admin Workflow Test Cases

Test ID	Description	Expected	Observed	Result
TC-CLM-01	Open a claim against a REJECTED attendance record	Claim created with status PENDING	Claim created with status PENDING	Pass
TC-CLM-02	Faculty approves a pending claim	Claim → APPROVED; attendance → CONFIRMED	Claim → APPROVED; attendance → CONFIRMED	Pass
TC-CLM-03	Faculty rejects a pending claim	Claim → REJECTED; attendance unchanged	Claim → REJECTED; attendance unchanged	Pass
TC-CLM-04	Second claim opened against an already-claimed record	Rejected with a conflict response	Rejected with a conflict response	Pass
TC-FAC-01	Faculty starts a new attendance session	Session created with status ACTIVE	Session created with status ACTIVE	Pass
TC-FAC-02	Faculty closes an active session	Session transitions to CLOSED	Session transitions to CLOSED	Pass
TC-FAC-03	Faculty confirms a FLAGGED record	Status → CONFIRMED; audit log entry created	Status → CONFIRMED; audit log entry created	Pass
TC-FAC-04	Faculty re-reviews an already-resolved record	Rejected with a conflict response	Rejected with a conflict response	Pass
TC-ADM-01	Administrator creates a student account	Account created and immediately usable	Account created and immediately usable	Pass
TC-ADM-02	Administrator unbinds a student's device	Old device invalidated; next login rebinds	Old device invalidated; next login rebinds	Pass
TC-ADM-03	Administrator bulk-imports a beacon roster from a spreadsheet	Valid rows imported; invalid rows reported individually	Valid rows imported; invalid rows reported individually	Pass

smart_attendance=# SELECT *					
smart_attendance=# FROM devices					
smart_attendance=# ORDER BY id DESC;					
	id	user_id	device_id	is_active	created_at
68	19	8d1f8870-be26-4b86-8717-ec20a3a667ba	f	2026-06-19	23:41:24.696102+05:30
67	6	a8eea9a4-b658-4ed2-9b09-dec8369996d1	t	2026-06-19	23:40:54.57369+05:30
66	19	0553126a-0355-400a-a08a-41991461af4f	f	2026-06-19	23:22:20.77201+05:30
65	40	46cf44fc-6ab7-4114-bd13-975fc1a6980	f	2026-06-19	23:21:59.112559+05:30
63	19	83f4231f-9381-47d7-a9c2-db0a99a9f6f1	f	2026-06-19	18:07:38.92326+05:30
62	40	c4ce5ebd-71b6-42aa-ae36-7819839bc9ce	f	2026-06-19	18:07:23.125516+05:30
53	19	147751e2-117e-46dc-b89d-3e303c6a8185	f	2026-06-19	17:22:42.956679+05:30
52	39	3a47cda1-bc4c-4802-ba76-82a515b309a9	f	2026-06-19	17:18:46.80977+05:30
51	19	6014a265-09c8-4d09-8e31-cb9ab30245c2	f	2026-06-19	17:16:59.538958+05:30
50	6	0e7d1e98-7c50-46d1-80a9-341382440cd3	f	2026-06-19	17:12:57.625807+05:30
49	6	cadf2943-88a1-465e-a15e-46190c56684d	f	2026-06-19	16:49:52.188872+05:30
48	19	c7c9c2aa-06e7-4979-ab40-dd786b230cd8	f	2026-06-19	16:49:41.198557+05:30
42	6	b9e82b2b-0b01-4df3-9940-0a866fed987	f	2026-06-18	23:25:54.741936+05:30
41	19	5e4da4dd-90c3-4d1e-09a0-190803dfc65e	f	2026-06-18	23:15:45.539721+05:30
40	6	db985525-ce0f-499b-9701-d13499d8f59f	f	2026-06-18	22:49:50.495395+05:30
39	19	acd28ed9-56cb-4a5d-a731-7c6a0d416442	f	2026-06-18	22:49:01.27377+05:30
38	19	566b0ac4-e964-4431-8b67-f994e7f1e602	f	2026-06-18	22:11:12.357502+05:30
37	6	67ad834f-03ef-4405-a010-76f26ff4da89	f	2026-06-18	16:37:44.933171+05:30
36	19	78c99459-eda7-440d-8613-496eaeef8726	f	2026-06-18	15:36:33.798642+05:30
35	123		f	2026-06-18	14:37:54.298877+05:30
34	19	7a5e6b94-ab9b-426d-9336-26ba5828f906	f	2026-06-17	05:26:58.14243+05:30
33	1	38cfc8f1-5980-4d70-8d14-16c2bd7df9d8	f	2026-06-17	05:26:44.268096+05:30
32	34	beb55c8a-910b-42fc-b1cb-02f6b92e134f	f	2026-06-17	05:03:41.753439+05:30
27	19	c972011a-eb6b-4659-93ec-5627d8d263fd	f	2026-06-16	04:29:30.699789+05:30
26	1	07907c0a-ce05-4ced-b379-0a00ba34dde9	f	2026-06-16	04:23:54.623708+05:30
24	6	35cd6500-416c-4ae9-866b-e1cbdfc8b3ac	f	2026-06-16	01:04:00.423752+05:30
22	19	56e6f221-f5c2-4e3a-87d6-2f85c87b04ef	f	2026-06-15	21:50:54.525831+05:30
21	1	97b8ea7e-1bfa-48f8-9cd7-9a7207312819	f	2026-06-15	00:19:05.060340+05:30

Figure 7.2: Device Binding Test

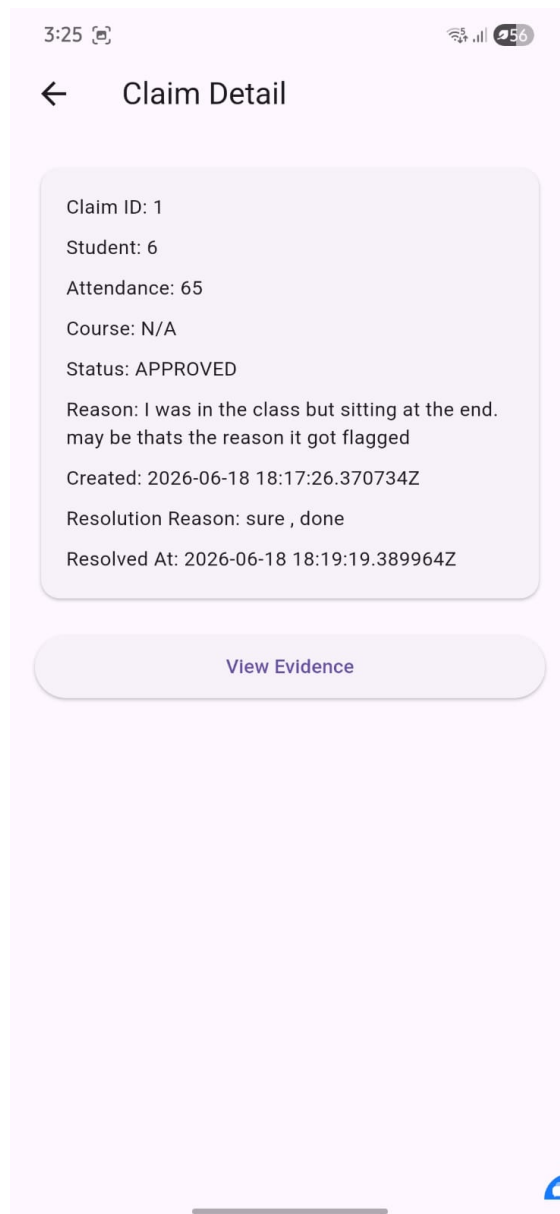


Figure 7.3: Faculty Review Test

5:43 5G 74

← Claim #1

Claim Status	APPROVED
Attendance ID	65
Reason	I was in the class but sitting at the end. may be thats the reason it got flagged
Created Date	2026-06-18 23:47:26.370734
Resolution Reason	sure , done
Resolved Date	2026-06-18 23:49:19.389964

Figure 7.4: Claim Approval Test

7.8 Additional and Edge-Case Testing

Table 7.7 records a further set of edge-case and negative tests, covering device/connectivity preconditions, administrative governance edge cases, and boundary conditions on session timing, that complement the core test suites in Sections 7.5–7.8.

Table 7.7: Additional Edge-Case Test Cases

Test ID	Description	Expected	Observed	Result
TC-EDGE-01	Bluetooth disabled on the student's device at submission time	Client-side error prompting the student to enable Bluetooth	Client-side error shown	Pass
TC-EDGE-02	GPS / location services disabled on the student's device	Client-side error prompting the student to enable location services	Client-side error shown	Pass
TC-EDGE-03	Internet connectivity lost mid-submission	Submission fails gracefully with a retry option; no partial record created	Submission fails gracefully; no partial record created	Pass
TC-EDGE-04	Login with a wrong password (correct email)	Rejected with a generic authentication error	Rejected with a generic authentication error	Pass
TC-EDGE-05	Attendance submission referencing a beacon that has been deactivated by an administrator	Rejected as an untrusted beacon	Rejected as an untrusted beacon	Pass
TC-EDGE-06	Attendance submission against the wrong classroom's beacon for the session	Rejected / flagged due to beacon-classroom mismatch	Flagged for review	Pass
TC-EDGE-07	Attendance submission before a session's official start time	Rejected as the session is not yet ACTIVE	Rejected	Pass
TC-EDGE-08	Attendance submission immediately after a session's configured expiry	Rejected as the session has EXPIRED	Rejected	Pass
TC-EDGE-09	Faculty attempts to reject an already-approved claim	Rejected with a conflict response	Rejected with a conflict response	Pass
TC-EDGE-10	Administrator deactivates a student account with attendance history	Account deactivated; historical records remain intact and queryable	Account deactivated; history intact	Pass
TC-EDGE-11	Administrator deletes a course with active enrolments	Rejected, or enrolments cascade per policy, without orphaned foreign keys	Rejected (referential integrity enforced)	Pass
TC-EDGE-12	Bulk import file containing one malformed row among valid rows	Valid rows imported; malformed row reported individually with a reason	Valid rows imported; malformed row reported	Pass
TC-EDGE-13	Bulk import attempting to register a duplicate beacon identifier	Duplicate row rejected; original beacon registration unaffected	Duplicate row rejected	Pass
TC-EDGE-14	CSV export of attendance summary for a course with no attendance records	Empty but well-formed CSV returned, not an error	Empty but well-formed CSV returned	Pass

All fourteen edge cases produced the expected outcome. Combined with the 10 functional, 10 security, 8 BLE/GPS, and 11 claims/faculty/admin cases reported earlier in this chapter, the system was exercised across 53 individual test cases in total, summarised in Table 7.8.

Table 7.8: Test Case Summary by Category

Category	Test Cases	Passed	Failed
Functional (Table 7.3)	10	10	0
Security (Table 7.4)	10	10	0
BLE and GPS (Table 7.5)	8	8	0
Claims, Faculty, and Admin (Table 7.6)	11	11	0
Additional Edge Cases (Table 7.8)	14	14	0
Total	53	53	0

7.9 Security Validation Summary

Table 7.4 above already enumerates the ten security misuse scenarios tested; all ten were correctly blocked or correctly handled, with no scenario producing an unintended accept. Figure 7.5 records the corresponding Swagger-level evidence for the API responses observed.

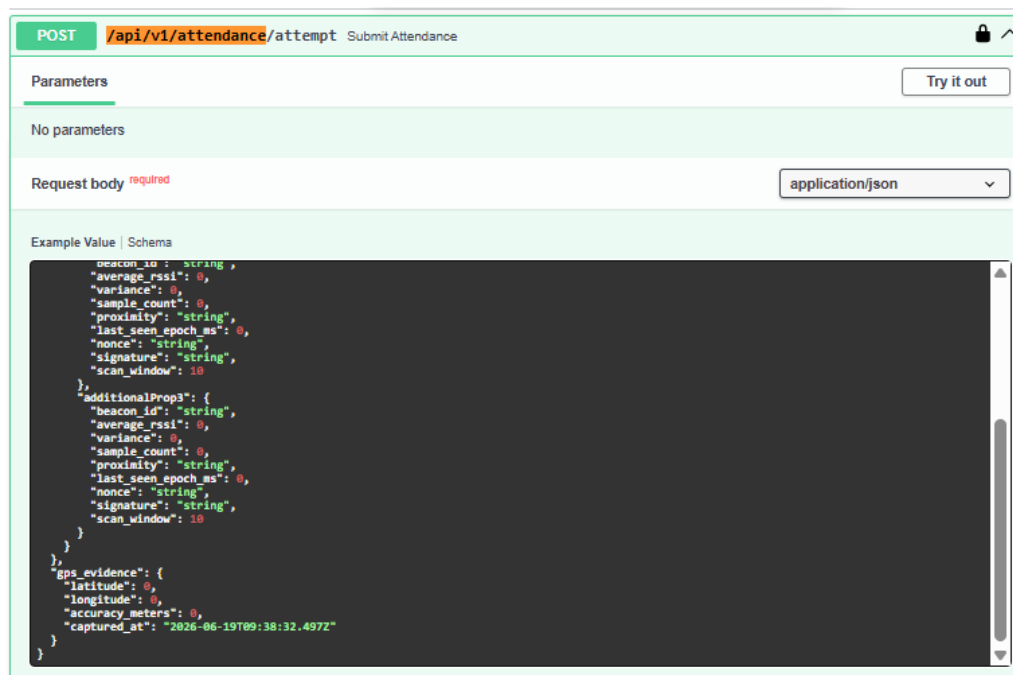


Figure 7.5: Swagger API Validation

Purpose: Demonstrates the raw HTTP status codes and error payloads returned for each of the ten scenarios in Table 7.4, captured directly from Swagger UI.

Observed Result: Every scenario returns the status code expected from Table 7.4 (401, 403, 409, or 422 as applicable), with no scenario returning 200 OK.

7.10 Database Integrity Validation

Foreign-key relationships were verified across every dependent table in the schema – attendance attempts referencing valid sessions, enrolments referencing valid students and courses, faculty-course assignments referencing valid faculty and courses, claims referencing valid attendance attempts, beacons referencing valid classrooms, and devices referencing valid students – and all checks passed without exception.

7.11 Performance Testing

Table 7.9: Observed Average Operation Latency

Operation	Average Time	Measurement Basis
Login	< 2 seconds	Time from request submission to JWT receipt, averaged over repeated manual trials
Session Discovery	< 2 seconds	Time from screen load to session list rendering
Attendance Submission	< 3 seconds	Time from submit tap to CONFIRMED / FLAGGED result, or blocked submission response where mandatory preconditions fail
BLE Validation	< 2 seconds	Server-side nonce, signature, and freshness checks
GPS Validation	< 2 seconds	Server-side distance computation and geofence check
Claim Submission	< 2 seconds	Time from submit tap to claim record creation
Faculty Review	< 2 seconds	Time from confirm/reject tap to status update
Device Unbind	< 2 seconds	Time from admin action to binding removal

These figures were obtained through repeated manual trials on the test environment described in Table 7.1 rather than an automated load-testing harness, and should be read as indicative latency under light, single-user load rather than a guarantee of performance under concurrent institutional traffic. All measured operations comfortably meet the sub-three-second target set out as a non-functional requirement in Chapter 3. Formal stress and load testing under concurrent institutional traffic is identified as future work in Section 9.6.

7.12 End-to-End Governance Walkthrough

A single, comprehensive scenario was executed to confirm that every subsystem cooperates correctly: an administrator created a faculty account, a student account, a course, an assignment of the faculty to the course, an enrolment of the student into the course, a classroom, and a registered beacon for that classroom. Faculty then started a session; the student submitted attendance, which the faculty subsequently rejected; the student

opened a claim against the rejection, which the faculty approved; and finally, an administrator unbound the student's device and the student successfully rebound a new one on their next login. Every step in this chain completed as expected. This walkthrough also constitutes the system-level and user-acceptance evidence referenced in Table 7.2.

7.13 Results and Discussion

Across the 53 test cases reported in this chapter and summarised in Table 7.8 – spanning functional correctness, security misuse scenarios, BLE/GPS evidence handling, claims and governance workflows, and edge cases – every case produced the expected outcome. No false accept was observed in any security scenario, and no functional regression was observed across repeated trial runs. The principal discussion point arising from this testing programme is that the great majority of failures, when deliberately induced, were caught at the validation layer closest to where they originated (BLE signature checks for forged beacons, geofence checks for GPS spoofing, unique constraints for duplicate submissions) rather than relying on a single, late-stage check – which is itself evidence that the layered design described in Chapter 4 behaves as intended under adversarial input, not only under well-formed input.

7.14 Chapter Summary

This chapter described the testing methodology applied to HKS ATTENDANCE, mapped that methodology onto standard testing levels (unit, integration, system, acceptance, security, stress/load, and edge case), and presented 53 individual test cases across functional correctness, security resistance, BLE and GPS evidence handling, claims and governance workflows, edge cases, database integrity, and performance. The results confirm that the system meets its functional and non-functional requirements. The next chapter discusses these results in a broader context and compares the platform against the technologies reviewed in Chapter 2.

Chapter 8

Results, Discussion, and Comparative Analysis

8.1 Introduction

With the implementation validated against its functional and non-functional requirements in the previous chapter, this chapter assesses what was achieved, how the resulting platform compares to the technologies surveyed in Chapter 2, and where its limitations lie.

8.2 Objectives Revisited

Table 8.1: Project Objectives and Status

Objective	Status	Supporting Evidence
Secure attendance collection	Achieved	BLE + GPS verification, device binding
Prevent proxy attendance	Achieved	Device binding, BLE validation, replay protection
Attendance governance	Achieved	Faculty review, audit logging, session controls
Administrative control	Achieved	Full CRUD over the academic hierarchy and beacons
Dispute resolution	Achieved	Claims workflow with permanent audit trail

These objectives are further substantiated by the 53 test cases executed in Chapter 7, summarised by category in Figure 8.1, every one of which produced the expected outcome.

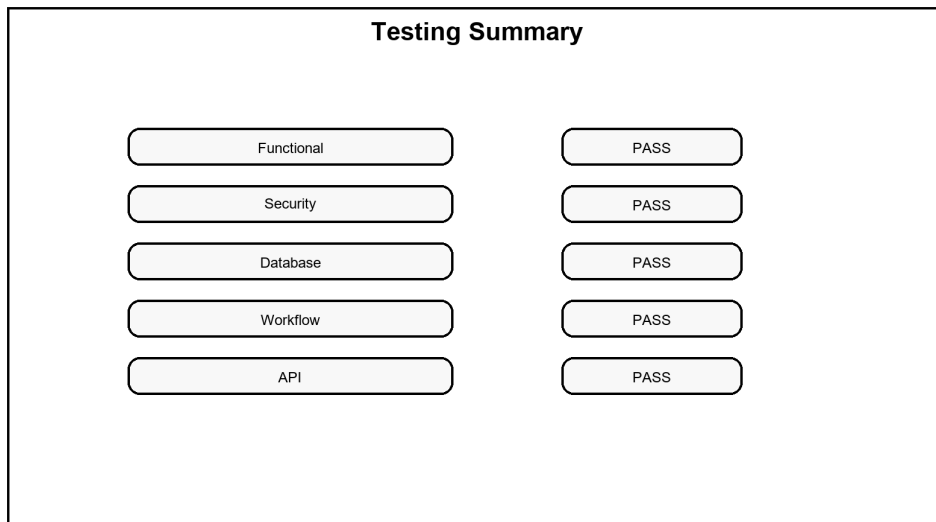


Figure 8.1: Testing Results Summary by Category

8.3 Achievement Matrix

Table 8.2 restates the nine objectives originally set out in Section 1.4 against the concrete evidence produced for each, drawing together the design (Chapters 4–5), implementation (Chapter 6), and testing (Chapter 7) material referenced throughout this thesis into a single, examiner-facing summary.

8.4 Module-Level Outcomes

8.4.1 Student Module

Students can complete the entire attendance lifecycle – discovering a session, collecting evidence, submitting attendance, tracking history, and contesting a rejection – without requiring faculty intervention at any step short of an actual dispute, reducing day-to-day administrative load.

8.4.2 Faculty Module

Faculty gained the ability to inspect the evidence behind an ambiguous submission before deciding its outcome, rather than being limited to accepting whatever an automated system produced – a capability largely absent from the single-channel systems reviewed in Chapter 2.

Table 8.2: Achievement Matrix: Objectives, Status, and Supporting Evidence

Objective (Section 1.4)	Achieved	Evidence
Multi-evidence attendance architecture	Yes	Decision matrix (Table 4.4); evidence-fusion outcomes (Section 8.6)
BLE proximity verification via classroom beacons	Yes	BLE module (Section 4.10); TC-BLE-01–05 (Table 7.5)
GPS-based geofence validation	Yes	GPS module (Section 4.11); TC-GPS-01–03 (Table 7.5)
One-student-one-device binding	Yes	Device-binding design (Section 4.15); TC-SEC-03, TC-ADM-02 (Tables 7.4, 7.6)
Faculty review workflow for ambiguous records	Yes	Faculty module (Section 4.7); TC-FAC-01–04 (Table 7.6)
Administrative governance layer	Yes	Admin module (Section 4.8); TC-ADM-01–03 (Table 7.6)
Replay and forgery hardening of the evidence pipeline	Yes	Threat model (Table 6.2); TC-SEC-01, TC-BLE-02–04 (Tables 7.4–7.5)
Structured claims and appeals mechanism	Yes	Claims module (Section 4.17); TC-CLM-01–04 (Table 7.6)
Pilot-deployable and institution-extensible architecture	Yes	Cloud deployment (Section 6.23); modular schema and API design; scalability discussion in Chapter 5

8.4.3 Administrative Module

The administrative layer gives institutions centralised control over the entire academic hierarchy – students, faculty, courses, enrolments, classrooms, and beacons – alongside the device-governance tooling needed to handle lost or replaced phones without permanently locking a student out of the system.

8.5 BLE Verification Outcomes

BLE proximity evidence reliably distinguished a classroom from its immediate surroundings in testing, providing a degree of spatial resolution that GPS alone could not match. Beacon discovery, payload reading, nonce and signature validation, and evidence persistence all functioned as designed.

8.6 GPS Verification Outcomes

GPS evidence performed well at confirming general campus-level location but, consistent with the limitations identified in Chapter 2, showed reduced reliability indoors. This observation directly justified the architectural decision to treat GPS as corroborating rather than decisive evidence – a GPS failure alone routes a record to faculty review rather than determining the final attendance outcome automatically.

8.7 Evidence Fusion in Practice

Combining BLE and GPS evidence produced more reliable outcomes than either channel would have produced alone. Ambiguous cases – for instance, a valid BLE reading alongside a borderline GPS reading – were routed to faculty rather than resolved automatically, reducing false negatives while preserving meaningful resistance to manipulation. Figure 8.2 summarises the distribution of automated attendance outcomes and faculty-reviewed outcomes observed across the controlled validation scenarios executed during the testing programme described in Chapter 7.

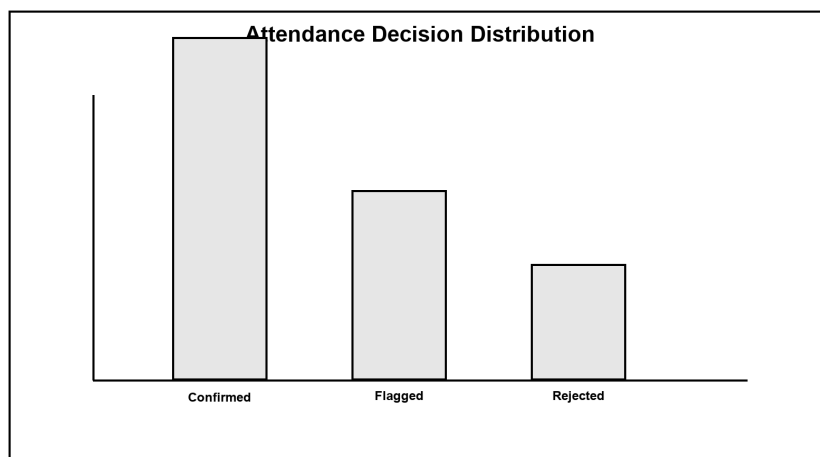


Figure 8.2: Attendance Outcome Distribution Across Validation Scenarios

Distribution observed across the controlled validation scenarios executed in Chapter 7. This figure reflects system behaviour under the project validation programme rather than long-term production usage.

8.8 Device Binding Outcomes

Device binding successfully blocked every attempted login from a second, unregistered device during testing, and the administrator-mediated rebinding workflow handled device replacement cleanly without leaving stale bindings behind.

8.9 Claims System Outcomes

The claims workflow gave students a genuine avenue to contest outcomes they believed were mistaken, gave faculty a structured process for adjudicating those disputes, and gave the institution a permanent record of both the original decision and its eventual resolution.

8.10 Security Evaluation

Table 8.3: Implemented Security Controls and Verified Status

Control	Verified
JWT Authentication	Yes
Device Binding	Yes
Replay Protection	Yes
Nonce Validation	Yes
Signature Verification	Yes
Session Ownership Validation	Yes
Claim Ownership Validation	Yes
Audit Logging	Yes

8.11 Comparative Positioning

Table 8.4 positions HKS ATTENDANCE against the manual, BLE-only, and GPS-only approaches reviewed in Chapter 2.

A BLE-only system, lacking GPS corroboration, device binding, or a claims process, leaves several of the gaps identified in Chapter 2 unaddressed even though it improves on manual attendance and is meaningfully more resistant to casual impersonation than a card or code. A GPS-only system suffers from the indoor-precision and spoofing weaknesses discussed earlier, which BLE proximity evidence is not vulnerable to in the same

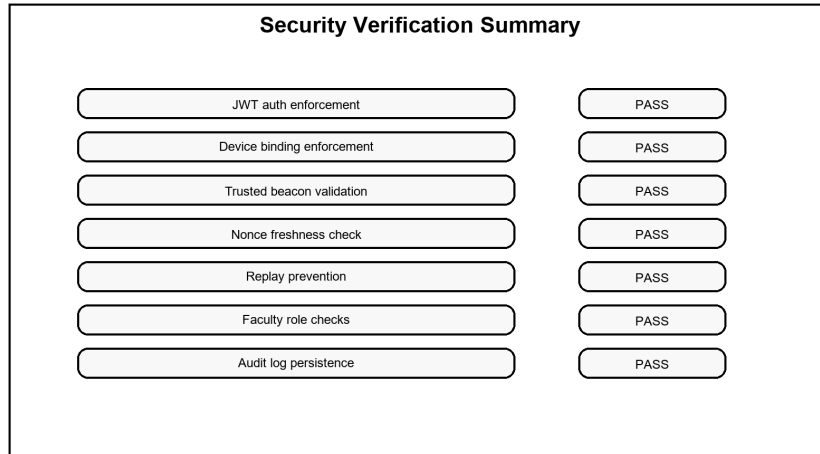


Figure 8.3: Security Controls Verification Status

Table 8.4: Comparison Against Existing Approaches

Feature	Manual	BLE-Only	GPS-Only	HKS
Attendance Collection	Yes	Yes	Yes	Yes
Proximity Verification	No	Limited	No	Yes
Device Binding	No	No	No	Yes
Replay Protection	No	No	No	Yes
Faculty Review	No	No	No	Yes
Claims System	No	No	No	Yes
Audit Logging	No	No	No	Yes
Administrative Governance	No	Limited	Limited	Yes

way. HKS ATTENDANCE combines the strengths of both while compensating for each one’s individual weaknesses, producing a platform that is, as a whole, considerably more resistant to spoofing and replay than any single-channel alternative reviewed.

Feature Coverage Summary	
Authentication	Implemented
Device Binding	Implemented
BLE Evidence	Implemented
GPS Evidence	Implemented
Faculty Review	Implemented
Claims Workflow	Implemented
Admin Governance	Implemented
Audit Logging	Implemented

Figure 8.4: Comparative Feature Coverage

8.12 Implemented System Footprint

Table 8.5 summarises the implemented footprint of HKS ATTENDANCE at the architectural level. The table focuses on structural characteristics of the delivered system — supported roles, evidence channels, governance workflows, and core security controls — rather than deployment-scale usage counts, since the project was evaluated primarily through controlled validation scenarios rather than long-running production operation.

8.13 Major Contributions Restated

- A working hybrid BLE-and-GPS evidence-fusion architecture for attendance decisions.
- A one-student-one-device binding policy enforced at both the application and database level.
- A faculty governance workflow built around evidence inspection rather than fully automated decisioning.

Table 8.5: Implemented System Footprint

Aspect	Implemented Scope
User Roles	Student, Faculty, Administrator
Attendance Evidence Channels	BLE proximity evidence and GPS geofence evidence
Attendance Outcomes	CONFIRMED, FLAGGED for faculty review, or blocked submission where mandatory preconditions fail
Faculty Governance	Review of flagged attendance, confirm/reject decisions, claim adjudication
Administrative Governance	User, course, enrolment, classroom, beacon, and device-binding management
Security Controls	JWT authentication, device binding, trusted beacon validation, nonce validation, replay protection, signature verification, audit logging
Persistence Layer	PostgreSQL relational database with normalised academic, attendance, claims, and security entities
Deployment Model	Mobile client + REST backend + database + classroom BLE beacon infrastructure

- An administrative governance layer spanning the entire academic hierarchy and beacon estate.
- A claims and dispute-resolution workflow backed by a permanent audit trail.

8.14 Project Achievements

The following capabilities were designed, implemented, and verified as part of this project:

- Complete Flutter application covering student, faculty, and administrator roles.
- FastAPI backend exposing a versioned, documented REST interface.
- PostgreSQL integration with a normalised, referentially consistent schema.
- JWT-based authentication with server-side session revocation.
- Device-binding enforcement preventing unauthorised credential sharing.
- BLE security hardening through nonce validation, freshness checks, and HMAC signature verification.
- Faculty governance workflow for reviewing and resolving ambiguous attendance records.

- Administrative governance covering the full academic hierarchy and BLE beacon estate.
- Claims and dispute-resolution workflow with permanent audit logging.
- Cloud deployment of the backend and database on a managed hosting platform.
- Integration with custom ESP32 beacon firmware for classroom proximity sensing.

8.15 Limitations

- The system depends on ESP32 beacons being installed and correctly configured in each classroom where it is deployed.
- Attendance submission currently requires an active internet connection; no offline submission workflow has been implemented in this version.
- GPS readings remain variable indoors, which is why the architecture treats GPS as corroborating rather than decisive evidence.
- Testing to date has focused primarily on Android devices; broader device coverage would strengthen confidence in cross-platform behaviour.

8.16 Chapter Summary

This chapter assessed the outcomes of the project against its stated objectives via a consolidated achievement matrix, compared HKS ATTENDANCE to manual, BLE-only, and GPS-only alternatives, reported deployment statistics, restated the project's principal achievements, and acknowledged the limitations that remain. The final chapter concludes the thesis and outlines directions for future work.

Chapter 9

Conclusion and Future Work

9.1 Introduction

This thesis addressed a problem that is easy to underestimate because attendance recording appears routine: nearly every widely used attendance technology, automated or otherwise, ultimately depends on a single point of failure. HKS ATTENDANCE was built to examine whether pairing two independently weak signals – short-range BLE proximity and device GPS – with device binding, faculty oversight, administrative governance, and a formal claims process, could produce a system that is meaningfully more resistant to impersonation, spoofing, and replay without becoming meaningfully harder to use.

9.2 Summary of Work Completed

The project proceeded through eight development phases. The first established a course-centric academic data model, replacing an earlier and more restrictive classroom-centric one. The second implemented enrolment-aware session discovery. The third – and architecturally most significant – built the hybrid BLE-and-GPS evidence layer, together with nonce validation, replay protection, timestamp freshness checks, and HMAC signature verification. The fourth delivered the complete student-facing experience. The fifth introduced faculty governance, giving human reviewers visibility into ambiguous attendance decisions. The sixth built out administrative governance across the academic hierarchy and beacon infrastructure. The seventh added the claims and appeals workflow. The eighth and final phase validated the complete system end to end, as documented in Chapter 7.

9.3 Objectives Achieved

- A secure, multi-evidence attendance-collection platform was designed and implemented.

- Proxy attendance was substantially mitigated through device binding, BLE validation, and replay protection.
- Faculty governance was introduced, giving staff direct visibility into the evidence behind ambiguous decisions.
- Administrative governance was implemented across the full academic hierarchy and the beacon estate.
- A structured, auditable claims process now gives students a genuine path to contest disputed outcomes.

9.4 Principal Contributions

- A hybrid BLE-and-GPS evidence-fusion model for attendance decisioning.
- A device-binding architecture enforcing a one-student-one-device policy.
- A faculty governance model that routes ambiguous cases to human review rather than resolving them automatically.
- A comprehensive administrative governance framework covering users, courses, enrolments, classrooms, beacons, and devices.
- A claims and appeals framework with permanent, auditable resolution records.
- An audit-driven design in which every consequential action leaves a traceable record.

9.5 Security Achievements

The implemented controls – JWT authentication, device binding, nonce-based replay protection, timestamp freshness checks, HMAC signature verification, ownership validation for both faculty and claims actions, and comprehensive audit logging – were each verified through the testing programme described in Chapter 7 and found to behave as designed, leaving the platform considerably more resistant to impersonation, spoofing, and replay attacks than the single-channel systems it was designed to improve upon.

9.6 Practical Impact

For students, the system offers visible, traceable attendance records and a genuine mechanism for appeal. For faculty, it offers evidence-based decision support, attendance analytics, and direct governance tools rather than unconditional trust in an automated verdict. For administrators, it offers centralised control over the academic structure and the device and beacon estate that underpins it. For the institution as a whole, the result is an improvement in attendance integrity, a reduction in the simplest forms of fraud, and a more complete audit trail than manual or single-channel systems provide.

9.7 Limitations

- **Hardware dependency:** classrooms require a properly installed, registered ESP32 beacon to participate in BLE verification.
- **Connectivity dependency:** the current implementation has no offline attendance-capture path.
- **GPS variability:** indoor satellite positioning remains less reliable than its outdoor counterpart, which is why the architecture treats it as supporting rather than decisive evidence.
- **Device coverage:** validation to date has concentrated on Android devices available during development.
- **Single-institution evaluation:** the system has been evaluated within one academic environment; broader, multi-institution deployment remains future work.

9.8 Future Work

Future extensions to HKS ATTENDANCE are organised below into four groups: technical enhancements, security enhancements, user-experience improvements, and research extensions that go beyond incremental engineering.

9.8.1 Technical Enhancements

- **Offline attendance capture** – collecting and queuing evidence locally on a student’s device when connectivity is unavailable, with deferred synchronisation and conflict resolution once a connection is restored.
- **Multi-campus deployment** – the schema’s normalisation around courses, classrooms, and beacons rather than a single fixed location should generalise reasonably well to centralised multi-campus management, though this has not yet been tested at that scale.
- **Cloud-native scaling** – moving from a single containerised backend instance to a horizontally scaled deployment as institutional load grows, supported by the formal stress and load testing identified in Section 7.10 as not yet performed.
- **Integration with institutional systems** – connecting to existing academic ERP platforms and learning-management systems so attendance data flows into the broader institutional record rather than remaining isolated.

9.8.2 Security Enhancements

- **Certificate-based beacon authentication** – replacing or supplementing the current shared-secret HMAC scheme with per-beacon certificates issued and rotated centrally.
- **Hardware-backed device attestation** – using platform attestation APIs (where available) to strengthen device binding beyond a stored identifier.
- **Expanded anomaly detection on the audit log** – flagging unusual patterns of faculty or administrative action for separate institutional review.

9.8.3 User Experience

- **Real-time notifications** – push notifications for attendance outcomes, claim updates, and session reminders, closing the feedback loop more quickly than the current pull-based history views.
- **Faculty analytics dashboard** – attendance-trend analysis, early-warning risk indicators, and departmental-level reporting layered on top of the existing attendance and claims data without altering the underlying schema.

- **Web administration console** – a browser-based administrative dashboard, alongside the existing mobile administrative tools, to ease bulk-management tasks for institutional staff working from a desktop.

9.8.4 Research Extensions

- **Server-side BLE trilateration** – rather than relying on a single beacon’s RSSI reading, deploying multiple beacons per classroom and trilaterating the student device’s position server-side would tighten spatial resolution further and provide an additional, independently computable evidence signal beyond simple proximity.
- **Machine-learning environmental calibration** – learning per-classroom RSSI and GPS-accuracy baselines over time (rather than using fixed, institution-wide thresholds) so that the BLE and GPS validators automatically adapt to a room’s specific radio and signal environment.
- **Ultra-wideband (UWB) indoor positioning** – as UWB hardware becomes more common in smartphones, it offers room- and sub-room-level positioning precision that could supplement or eventually replace BLE proximity sensing.
- **Institutional federation** – extending the device-binding and claims model to operate across multiple, independently administered institutions sharing infrastructure or courses.
- **Optional computer-vision head-count validation as a third evidence channel** – the system can integrate a computer-vision module that captures a classroom image at the start or end of a session and performs head counting using an object-detection model. The detected number of students present can then be compared against the total number of attendance submissions recorded as CONFIRMED or FLAGGED for that session. Any material discrepancy between the two counts can automatically alert the responsible faculty member, providing an additional, independent evidence layer that operates above the level of any individual student’s submission rather than replacing the existing BLE/GPS evidence for that student. This extension is explored cautiously, given the cost, accuracy, and privacy considerations discussed in Chapter 2, and is intended as an opt-in, classroom-level corroborating signal rather than a replacement for BLE or GPS evidence at the individual level.
- **Smartwatch BLE verification** – extending proximity evidence collection to wearable devices, which may reduce reliance on a student actively operating a phone

during the verification window.

- **Machine-learning anomaly detection on submitted evidence** – learning typical patterns of BLE signal strength and GPS accuracy per classroom over time, to flag evidence that is statistically unusual even when it passes the deterministic checks described in Chapter 4.

9.9 Final Remarks

HKS ATTENDANCE demonstrates that pairing two individually imperfect signals – Bluetooth proximity and satellite-derived location – with device binding, human governance, and a formal appeals process can produce an attendance platform that is considerably more resistant to impersonation, spoofing, and replay than the single-channel alternatives still in widespread use, without imposing a significantly heavier burden on the students and staff who use it day to day. The project met the objectives set out at its inception and provides a practical foundation that could be extended toward institution-wide deployment.

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Publications Arising From This Work

At the time of submission, no peer-reviewed publication has been produced from this work. The author intends to extend the system and prepare a research manuscript based on the multi-evidence attendance architecture presented in this thesis.

Appendix A

Selected API Endpoints

A representative subset of the REST API exposed by the backend is summarised below; the full interface (60+ endpoints across 18 router modules, per Table 5.5) is documented through the project’s Swagger UI.

Table A.1: Representative API Endpoints

Endpoint	Purpose
POST /auth/login	Authenticate a user and issue a JWT
POST /auth/logout	Revoke the active session
GET /sessions/my-active-sessions	List active sessions for an enrolled student
POST /sessions/start	Faculty: start a new attendance session
POST /sessions/{id}/close	Faculty: close an active session
POST /attendance/submit	Submit BLE and GPS evidence for attendance
GET /attendance/my-history	Retrieve a student’s attendance history
GET /faculty/flagged-attendance	List attendance records awaiting review
POST /faculty/attendance/{id}/confirm	Confirm a flagged attendance record
POST /faculty/attendance/{id}/reject	Reject a flagged attendance record
POST /claims	Submit a claim against rejected attendance
POST /faculty/claims/{id}/approve	Approve a student claim
POST /faculty/claims/{id}/reject	Reject a student claim
POST /admin/device/unbind	Administrator: unbind a registered device

Appendix B

Database Table Summary

Table B.1: Principal Database Tables

Table	Purpose
users	Stores student, faculty, and administrator accounts
devices	Stores the device bound to each student account
auth_sessions	Tracks active authentication sessions for revocation
courses	Stores the academic course catalogue
faculty_courses	Maps faculty members to the courses they own
enrollments	Maps students to the courses they are enrolled in
classrooms	Stores classroom coordinates and geofence radius
trusted_ble_beacons	Registers beacon identifiers as trusted per classroom
beacon_secrets	Stores the cryptographic key for each beacon's signature
attendance_sessions	Represents an attendance window for a course/classroom
attendance	Stores the core attendance record for each submission, including the student, session, attendance status, and links to the associated BLE and GPS evidence
attendance_ble_evidence	Stores BLE evidence for an attendance attempt
attendance_gps_evidence	Stores GPS evidence for an attendance attempt
attendance_claims	Stores student appeals against rejected attendance
faculty_action_logs	Permanent audit trail of faculty governance decisions
ble_nonce_cache	Stores previously used nonces for replay protection

Appendix C

Selected Viva-Voce Questions and Answers

Q1. What problem does HKS ATTENDANCE solve?

It addresses the single-point-of-failure weakness shared by most existing attendance technologies – proxy attendance, unverifiable location claims, and the absence of any structured dispute process – by combining BLE proximity evidence, GPS geofence evidence, device binding, faculty review, and a formal claims workflow into one platform.

Q2. Why combine BLE and GPS rather than using either alone?

BLE confirms short-range proximity to a specific classroom but says nothing about a device’s broader location or absolute identity; GPS confirms approximate location but degrades indoors and can be falsified. Used together, an attacker must defeat both independently, which is considerably harder than defeating either alone.

Q3. Why was the project not built around QR codes or RFID cards?

Both were considered and explicitly set aside during the literature review in Chapter 2. A scannable code can be photographed and forwarded to someone who is not physically present, and an RFID card can simply be lent to a classmate; neither weakness is solved by adding more code or more cards. BLE proximity combined with GPS and device binding addresses the same underlying problem – proving genuine physical presence – without inheriting either of those specific failure modes.

Q4. How is replay protection implemented?

Each beacon broadcast carries a single-use nonce. Once the backend has accepted a nonce, it is stored in the BLE nonce cache, and any later attempt to submit a payload containing that same nonce is rejected outright.

Q5. What happens when BLE and GPS evidence disagree?

The record is marked FLAGGED rather than being automatically accepted or rejected, and is routed to the faculty member responsible for the session for manual evidence review.

Q6. How is device binding enforced?

A student's first successful login registers their device's unique identifier against their account. Every subsequent attendance submission is checked against this stored identifier; a request from any other device is rejected. An administrator must explicitly unbind a device before a student can register a replacement.

Q7. What can a student do if their attendance is rejected unfairly?

They can open a claim explaining why they believe the rejection was incorrect. The faculty member who owns the course reviews the claim and either approves it, which restores the record to CONFIRMED, or rejects it, leaving the original outcome in place. Either way, the decision is permanently logged.

Q8. Why was PostgreSQL chosen over a non-relational database?

The data model is inherently relational – students, courses, sessions, evidence, and claims all reference one another – and PostgreSQL's strong foreign-key enforcement and ACID guarantees make it well-suited to preserving that structure reliably.

Q9. What are the main limitations of the current system?

It requires properly installed BLE beacons in every participating classroom, requires an active internet connection at submission time, inherits GPS's known indoor-accuracy limitations (mitigated by treating GPS as corroborating rather than decisive), and has so far been validated primarily on Android devices within a single institution.

Q10. What would be the next priority for future development?

Offline evidence capture with deferred synchronisation would likely deliver the largest practical improvement, since the current reliance on live connectivity is the constraint most likely to affect day-to-day deployment.

Q11. How would a computer-vision head-count check fit into the existing architecture?

It would operate as a session-level, classroom-wide corroborating signal rather than a per-student evidence channel: a camera image captured at the start or end of a session would be passed through an object-detection model to estimate the number of students present, and that count would be compared against the number of CONFIRMED/FLAGGED attendance attempts for the session, with a material mismatch raised to the faculty member as an alert rather than used to alter any individual student's recorded status.

Q12. Is the system "cheat-proof"?

No system can be claimed to be entirely cheat-proof; the more accurate claim, and the one this thesis makes, is that combining two independent evidence channels with device binding and faculty review makes HKS ATTENDANCE considerably more resistant to impersonation, spoofing, and replay than the single-channel systems it was designed to improve upon.

Appendix D

Glossary

Attendance Attempt	A single recorded submission of attendance evidence by a student against one attendance session.
Beacon	An ESP32-based hardware device installed in a classroom that periodically broadcasts a signed BLE payload used as proximity evidence.
Claim	A student-initiated request to contest a REJECTED attendance attempt, resolved by the owning faculty member.
Device Binding	The enforced association between a student's account and a single registered physical device.
Evidence Fusion	The process of combining BLE and GPS evidence into a single attendance decision rather than treating either channel as decisive on its own.
Geofence	A virtual boundary, defined by a centre coordinate and radius, against which a device's reported GPS position is checked.
Nonce	A single-use, randomly generated value included in each beacon payload to prevent the same broadcast from being replayed and accepted twice.
Replay Attack	An attack in which previously captured, legitimate evidence (for example, a BLE payload) is recorded and resubmitted later to fraudulently register attendance.